

Delirium and long term cognition in critically ill patients

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ABSTRACT

Delirium, a form of acute brain dysfunction, is very common in the critically ill adult patient population. Although its pathophysiology is poorly understood, multiple factors associated with delirium have been identified, many of which are coincident with critical illness. To date, no drug or non-drug treatments have been shown to improve outcomes in patients with delirium. Clinical trials have provided a limited understanding of the contributions of multiple triggers and processes of intensive care unit (ICU) acquired delirium, making identification of therapies difficult. Delirium is independently associated with poor long term outcomes, including persistent cognitive impairment. A longer duration of delirium is associated with worse long term cognition after adjustment for age, education, pre-existing cognitive function, severity of illness, and exposure to sedatives. Interestingly, differences in prevalence are seen between ICU survivor populations, with survivors of acute respiratory distress syndrome experiencing higher rates of cognitive impairment at early follow-up compared with mixed ICU survivor populations. Although cognitive performance improves over time for some ICU survivors, impairment is persistent in others. Studies have so far been unable to identify patients at higher risk of long term cognitive impairment; this is an active area of scientific investigation.

Introduction

Delirium, a neuropsychiatric syndrome characterized by acute, fluctuating disturbances of consciousness and cognition, is the most common manifestation of critical illness related brain dysfunction, occurring in up to 80% of the sickest patients admitted to an intensive care unit (ICU).¹⁻⁵ Patients who need ICU admission are at risk for delirium because of the severity of their illness and exposures in the ICU (for example, sedation and immobilization), a danger that may be exacerbated by cerebral metabolic insufficiency, peripheral inflammation and/or neuroinflammation, and neurotransmitter imbalances, as well as neuroanatomical substrates and failure of network connectivity.^{6,7}

Delirium is a major public health problem, as it is associated with persistent morbidity including long term impairment in cognitive performance. Although the exact mechanism of experienced long term morbidities in ICU survivors is unknown, delirium is thought to accelerate cognitive decline in those with and without pre-existing impairment in cognitive function.⁸⁻¹¹ A recent meta-analysis of 24 studies (including 3562/6987 patients who experienced delirium) showed a significant association between delirium and long term cognitive impairment (estimated effect size (Hedges *g*) for 23

studies was 0.47 (95% confidence interval 0.35 to 0.59); $P < 0.001$).¹² Other long term consequences associated with delirium include restrictions in motor function,^{9,13} ongoing need for care in long term care institutions,^{14,15} a higher likelihood of discharge to destinations other than home,¹⁶ and greater long term disability.¹⁷

The most recent evidence based guidelines and consensus statements recommend the prioritization of a multicomponent non-drug approach to the management of delirium,¹⁸⁻²¹ with a focus on prevention and early recognition. Embedded within recent evidence based clinical practice guidelines are recommendations to implement care bundles that include use of a validated instrument for daily assessment of all ICU patients for delirium.²⁰⁻²⁴ Despite evidence of benefit, rates of compliance with care bundles remain low.²⁵ This shortcoming may be of particular importance given the recent covid-19 pandemic, during which standard of care has been strained by resource limitations (box 1). Recent reports from regions of the world hardest hit by covid-19 suggest that a flexible approach to management algorithms may be needed (box 2).

This review focuses on critical illness related delirium and long term cognitive impairment. It is aimed at clinicians caring for patients during and

Box 1: Acute neurologic manifestations in patients infected with covid-19

Early studies indicate that 20-30% of patients with covid-19 will present with or develop delirium during their hospital admission, with rates of 60-70% in cases of severe illness.²⁶ In a prospective cohort (n=140) of critically ill patients from two ICUs in Strasbourg, France, delirium was diagnosed in 80% of patients at least once during their ICU stay; hyperactive delirium was more common (87%) than hypoactive delirium (13%).²⁷ Experienced agitation required prolonged use of neuroleptic and sedative agents, which was associated with a statistically significant increase in the duration of mechanical ventilation (mean ratio 1.49; 95% confidence interval 1.01 to 2.20; $P=0.045$).²⁷ In a retrospective cohort (n=56), the use of low molecular weight heparin (enoxaparin 1 mg/kg/d) was associated with less frequent delirium events ($P=0.004$) and was accompanied by lower C reactive protein concentrations ($P=0.006$).²⁸ Heparin has been implicated in binding to SARS-CoV-2 spike proteins as well as down-regulating interleukin-6.²⁹ Interleukin-6 has been shown to be elevated in covid-19 patients.³⁰ Prevention of thrombosis and its subsequent inflammatory process may reduce the occurrence of delirium in covid-19 patients.²⁸ Further studies are ongoing.

In addition to direct structural changes being associated with delirium and other neurologic changes in patients with covid-19, patients with severe disease may receive deeper sedation, fewer breathing trials, and more limited mobility sessions than other critically ill patients. A worldwide two day prevalence study of ABCDEF* bundle practices across 212 ICUs in 38 countries found implementation rates to be low: pain assessment (45%), paired spontaneous awakening trial and spontaneous breathing trial (28%), sedation assessment (52%), delirium assessment (38%), early mobility and exercise (16% if patient being mechanically ventilated), and family engagement and empowerment (16%).³¹ In the face of staggering numbers of critically ill patients with covid-19 met with an imbalance of staffing and resources to meet the demand,³² patients are unlikely to receive the usual quality of care. Prioritizing minimization of sedation, daily breathing trials, early mobility, and other evidence based practices is time intensive and coordination intensive. The impact of unprecedented strain on ICU capacity during different peaks of the pandemic has yet to be determined; however, many critically ill patients may not have been receiving the same standard of clinical care as before the pandemic (for example, prolonged sedation), and as a result their outcomes, such as rates of delirium and death, may be worse.³³ Critically ill patients in isolation as a result of their disease are particularly vulnerable as covid-19 related concerns create further challenges to receipt of usual care. Most recently, a multicenter cohort study including 2088 patients (88% mechanically ventilated) from 69 ICUs across 14 countries reported a prevalence rate of a patient ever being delirious, over the 21 day data collection period, of 55%.³⁴ Requirement for mechanical ventilation; use of restraints; opioid, benzodiazepine, and vasopressin infusions; and antipsychotic administration were all associated with an increased risk of delirium the following day (all $P\leq 0.04$), whereas family visitation was associated with a decreased risk of delirium ($P<0.0001$).³⁴

*Assess, prevent, and manage pain, Both spontaneous awakening trials and spontaneous breathing trials, Choice of analgesia or sedation, Delirium: assess, prevent, and manage, Early mobility and exercise, and Family engagement and empowerment.

after ICU admission. We describe the epidemiology, criteria and tools for identification, and approaches to management of delirium in a critically ill patient population. We also describe the epidemiology, risk factors, and interventions for long term cognitive impairment in ICU survivors.

Sources and selection criteria

We did a PubMed search of articles published from 1 January 2010 to 1 September 2020, using the following search terms in combination with "delirium": "delirious", "acute confusion" or "inattention", "CAM-ICU" or "ICDSC", "management", "ICU", "acutely ill", or "critically ill". We

included only peer reviewed articles written in English and involving adults. We screened 6590 titles for relevance, selected 426 for more detailed review on the basis of title and abstract, and finally included 123 articles. We gave precedence to large key studies that have informed international guidelines. We also reviewed a wider range of recent studies for which conclusions and evidence are mixed. We excluded case reports and small case series of non-covid-19 ICU patients. We did, however, include articles of historical importance, including pivotal trials outside of the searched decade. In addition, we searched the reference lists of high quality articles and reviews, and we included selected randomized controlled trials (RCTs), observational studies, systematic reviews, and meta-analyses from these sources. In reviewing the covid-19 literature, we included lower quality of evidence, including case series and case reports, but maintained the requirement for peer review. The search for covid-19 specific literature was last updated on 31 December 2020. All studies were organized in EndNote X9 (fig 1).

Epidemiology

Delirium is a syndrome characterized by an acute change or fluctuation in baseline mental status, inattention, and either disorganized thinking or altered level of consciousness. The cardinal features of delirium are disturbed level of consciousness, with reduced ability to focus, sustain, or shift attention, and either a change in cognition or the development of a perceptual disturbance (that is, hallucinations).⁴¹ The classic three motoric presentations of delirium include hyperactive delirium (2%), characterized by agitation, aggression, disorientation, and hallucinations; hypoactive delirium (44%), characterized by lethargy, motor slowness, and social withdrawal; and mixed delirium (55%), characterized by fluctuations between hypoactive and hyperactive subtypes⁴²⁻⁴⁴ (table 1).

The prevalence of delirium in an ICU patient population ranges from 32% to 87%.⁴⁵⁻⁴⁹ Prevalence seems to vary considerably depending on whether the population studied received mechanical ventilation and other risk factors. For instance, the prevalence of delirium in a study of ventilated burn victims was 77%,⁴⁶ compared with 20% in a study of patients who were never intubated.⁴⁹ Different clinical phenotypes (phenotypes defined by clinical risk factors) of delirium were first used in an observational cohort of 97 patients receiving mechanical ventilation and showing differential responses to daily interruption of continuous sedation.⁵⁰ Delirium that was rapidly reversible with interruption of sedation was associated with shorter duration of mechanical ventilation, length of stay in ICU, and length of stay in hospital than was persistent delirium,⁵⁰ although the percentage of patients with rapidly reversible delirium was small (12%). Patients with persistent delirium were sicker, older, more septic, and likely more encephalopathic.⁵⁰ This study was the first to show the existence of different

Box 2: Long term cognitive outcomes of survivors of covid-19 related critical illness

Few data are available on long term outcomes and recovery after severe covid-19 infection. A meta-analysis of 28 studies (n=2820) assimilated data on long term outcomes after coronavirus infection (SARS and MERS), reporting high rates of post-traumatic stress (39%), depression (33%), and anxiety (30%) at six months.³⁵ A prospective cohort study of covid-19 survivors (n=143), 19 of whom needed ICU admission, found high rates of persistent symptoms of fatigue, dyspnea, joint pain, and chest pain for two months after hospital discharge.³⁶ Cognitive impairment was not studied in these investigations. However, in a prospective cohort study comparing outcomes between patients with covid-19 who had survived an admission to the ward and those who needed ICU admission, after a mean of 110 days following hospital discharge, 34% of patients complained of memory loss and no difference was found between groups in terms of perceived symptoms.³⁷ In a single center case series, 28 patients were seen six weeks after hospital discharge, and 57% had mild cognitive impairment on completing the Montreal Cognitive Assessment.³⁸ In one case report, a 39 year old patient transferred for inpatient physical rehabilitation for profound ICU acquired weakness was later determined to have moderate to severe cognitive impairment (processing speed, waking memory, visuospatial processing, and executive function).³⁹ As a result, a multimodal cognitive rehabilitation strategy was incorporated into his physical rehabilitation, demonstrating the diverse rehabilitation challenges considering likely occult and subtle cognitive changes in covid-19 survivors.³⁹ Reduced access to essential rehabilitation services exacerbates delirium and is likely to impede cognitive recovery after critical illness. Many outbreaks of SARS-CoV-2 infection have occurred in skilled facilities. Patients and their caregivers may be reluctant to accept placement at a post-acute care facility given the possible risks; if they are willing, limits or delay to access may be present owing to closures, downsizing, or processes in place for quarantine. As a result, a greater proportion of patients may be discharged home, placing additional responsibility on informal caregivers and outpatient services. In a recent prospective cohort study of covid-19 survivors (n=1648; 405 received treatment in the ICU), 78% of patients were discharged home and 13% were discharged to a skilled nursing or rehabilitation facility.⁴⁰

phenotypes with distinct outcomes. More recently, in a large multicenter, prospective cohort of 1040 ICU patients with acute respiratory failure, shock, or both, patients developed multiple clinical phenotypes of delirium including sedative associated, hypoxic, septic, metabolic, and unclassified delirium (that is, delirium in the absence of all other clinical risk factor phenotypes).⁵¹ During their ICU stay, more than half of patients experienced hypoxic, septic, or sedative associated delirium, whereas metabolic and unclassified phenotypes were less common.⁵¹ The percentage of patients experiencing either hypoxic or sedative associated delirium decreased over the course of ICU stay. Longer durations of sedative associated delirium, the most commonly experienced phenotype, predicted worse long term cognitive performance at both three and 12 months. Similarly, longer durations of hypoxic delirium (difference in score comparing three days versus zero days: -3.76, 95% confidence interval -7.16 to -0.37), septic delirium (-3.67, -7.13 to -0.22), and unclassified delirium (-4.70, -7.16 to -2.25) also predicted worse cognitive function at 12 months, whereas metabolic delirium did not.⁵¹ These findings would suggest that clinicians should consider different clinical phenotypes of delirium as distinct

indicators of acute brain injury and, in so doing, attempt to identify and mitigate relevant risk factors in an attempt to reduced long term sequelae (fig 2).

Effect of delirium on outcomes in ICU patients

In a recent meta-analysis, the risk ratio for death in patients with delirium was 2.2 (95% confidence interval 1.8 to 2.7; $P<0.001$).⁵² Patients with delirium have increased mortality at six months, compared with patients who were never delirious (approximately 20-25% increase in risk)^{49 53}; one prospective cohort study showed that the number days of delirium in the ICU was associated with time to death at one year after admission to ICU (hazard ratio 1.1, 1.0 to 1.2).⁵⁴ These findings are at odds with those of a recent, larger prospective cohort of more than 1000 ICU survivors that found no association between days delirious in the ICU and mortality at one year, after adjustment for sex, type of admission, Acute Physiology and Chronic Health Evaluation (APACHE) IV score, and the cumulative Sequential Organ Failure Assessment (SOFA) score throughout the ICU stay.⁵⁵ In a prospective cohort of similar size, again no association was seen between delirium and mortality after adjustment for time varying confounders, suggesting that the evolution of disease severity before the onset of delirium was responsible for increased risk of death in ICU.⁵⁶ An increased absolute risk of mortality of 2.0% (95% confidence interval 1.2% to 2.8%), however, was seen in post hoc analyses, when delirium persisted for at least two days.⁵⁶ Patients with delirium had longer length of stay (standard mean difference 1.4 (0.99 to 1.8) days; $P<0.001$) than did those without delirium. Additionally, mean duration of mechanical ventilation was approximately two days longer than in patients without delirium (standard mean difference 1.8 (0.31 to 3.3) days; $P<0.001$).⁵² Delirium carries an annual cost of \$4bn to \$16bn (£2.83-11.33bn; €3.28-13.13bn) in the US alone.⁵⁷

Assessment of delirium in the ICU

The Confusion Assessment Method for the ICU (CAM-ICU) and Intensive Care Delirium Screening Checklist (ICDSC) are the most commonly used tools for assessing delirium. On the basis of validation studies, they are recommended in most clinical practice guidelines, although other delirium assessment tools have been developed for use in the ICU (table 2). The CAM-ICU has been validated to assess delirium in ICU patients who may be non-verbal owing to mechanical ventilation and who can be assessed while sedated (unless they are comatose—that is, do not respond in any way to verbal stimulus).⁴⁸ Using a structured format that can be applied during a two to five minute examination, this tool evaluates four features of delirium—namely, acute onset or fluctuating course of mental status change, inattention, disorganized thinking, and altered level of consciousness. When administered by bedside nurses with no formal psychiatric training, the CAM-ICU showed high accuracy (sensitivity of 93-100%;

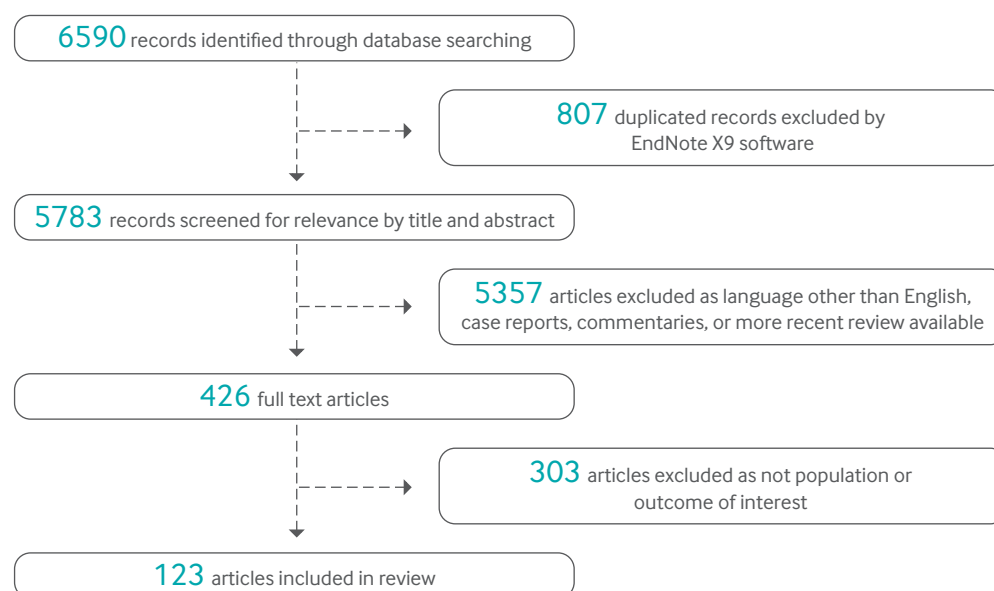


Fig 1 | Flow diagram of included articles

specificity 98-100%) and inter-rater reliability ($\kappa=0.96$) in a single center study.⁴⁸ The eight domain ICDSC, by contrast, assesses patients for eight signs and symptoms of delirium throughout a nursing shift, with a score of four or greater being positive for delirium. Interestingly, in a validation study of the ICDSC in medical-surgical ICU patients, the checklist showed a higher sensitivity (99%) than the CAM-ICU but a lower specificity (64%).⁶⁶ The items of the ICDSC (consciousness levels, inattentiveness, disorientation, hallucination, psychomotor agitation or retardation, inappropriate speech or mood, sleep-wake disturbance, symptom fluctuation) can be evaluated during daily routine nursing assessments.

Management of delirium

Risk factor reduction

The initial management of delirium should involve a detailed review of possible risk factors that may have triggered the episode of delirium. A systematic review of 33 studies identified 11 risk factors for delirium in the ICU, supported by medium to strong evidence.⁶⁷ Risk factors for delirium can be broadly categorized into three categories: features of the acute illness, patient or host related factors, and environmental or iatrogenic factors⁶⁸⁻⁶⁹ (fig 3). Acute illness or precipitating variables include coma, severity of illness (for example, measured by the APACHE II or SOFA score), emergency surgery, polytrauma, and mechanical ventilation; characteristics of patients that are related to a

higher risk of developing delirium include higher age and chronic pathology (for example, dementia, hypertension); most environmental or iatrogenic variables such as medication administration were inconclusive based on the available evidence at the time.⁶⁷⁻⁷⁰ Of the 11 more definitive risk factors identified in the systematic review,⁶⁷ the potentially modifiable ones included minimizing the duration of sedative associated coma,⁴⁶⁻⁷¹⁻⁷⁶ reducing days of mechanical ventilation,⁷⁴⁻⁷⁶⁻⁸⁰ and promoting the use of dexmedetomidine.⁸¹⁻⁸⁴ More recently, a systematic review of 20 studies found that neither age nor sex was associated with a delirium subtype.⁸⁵ Heterogeneity in study design and inconsistent reporting of subtype specific risk factors, however, limited data synthesis.⁸⁵

Non-drug therapies for delirium in the ICU

In an RCT (n=104), early physical and occupational therapy during spontaneous awakening trials in mechanically ventilated medical patients reduced the duration of delirium (median 2.0 (interquartile range 0.0-6.0) days versus 4.0 (2.0-8.0) days; $P=0.02$ in the intervention and usual care groups, respectively).⁸⁶ Similarly, early mobilization of surgical patients led to improved functional status and shortened duration of ICU stay.⁸⁷ In this multicenter RCT in 200 patients, early mobilization reduced duration of delirium (group difference 3.0 (95% confidence interval 0.5 to 5.5) days; $p=0.016$).⁸⁷ Furthermore, the benefit of early mobilization was consistent

Table 1 | Motor presentations of delirium

Subsyndromal delirium	Incomplete presentation of delirium syndrome
Hyperactive delirium	Characterized by motor agitation, restlessness, and aggressiveness
Hypoactive delirium	Characterized by motor retardation, apathy, and slowing of speech
Mixed hyperactive and hypoactive delirium	Characterized by mix of motor features
Catatonic retardation or catatonic excitement	Extreme forms of hypoactive and hyperactive delirium, respectively

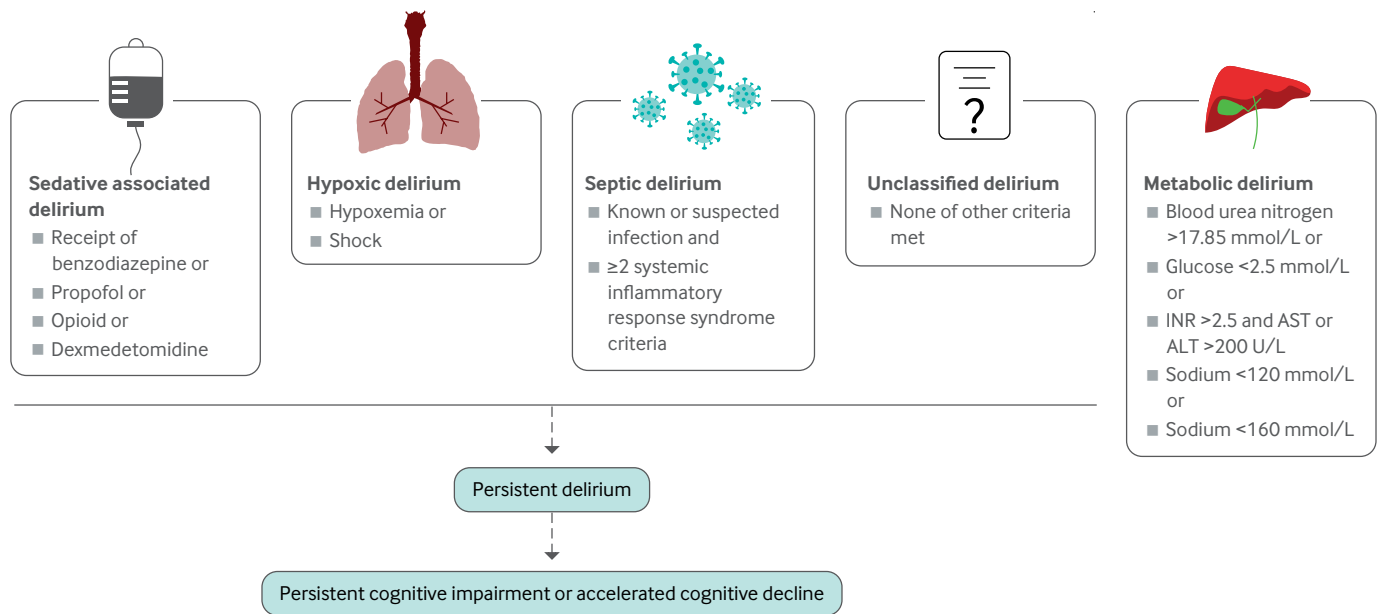


Fig 2 | Delirium phenotypes and risk of persistent cognitive impairment. ALT=alanine transaminase; AST=aspartate aminotransferase; INR=international normalized ratio

across Glasgow Coma Scale (GCS) levels without evidence of effect modification in a post hoc analysis for the primary outcome of functional independence at hospital discharge ($P=0.53$ for interaction).⁸⁸ Early mobilization is a component of evidence based care bundles for critically ill patients, promoting awake, spontaneously breathing, and mobile patients. Compliance with such bundles of care is associated with reduced risk of delirium on the next day (adjusted odds ratio 0.60, 0.49 to 0.72) and other clinically important outcomes.²²

Drug therapies for delirium in the ICU

Although beneficial, non-drug management is not always effective in preventing or treating delirium. As a result, several studies have investigated the effects of pharmacologic agents, particularly antipsychotic drugs, in the management of delirium in the ICU.

Antipsychotic agents

The largest trial of antipsychotic treatment to date, the Modifying the Impact of Neuropsychological Dysfunction-USA (MIND-USA) study, was a multisite RCT comparing haloperidol, ziprasidone, and placebo in 566 patients with delirium, which found no significant difference in number of days alive without delirium or coma during the 14 day intervention period.⁸⁹ The median number of days alive without delirium or coma was 8.5 (95% confidence interval 5.6 to 9.9) in the placebo group, 7.9 (4.4 to 9.6) in the haloperidol group, and 8.7 (5.9 to 10.0) in the ziprasidone group ($P=0.26$ for overall effect across trial groups).⁸⁹ Furthermore, no significant between group differences were seen in the secondary endpoints (30 day and 90 day survival, time to freedom from mechanical ventilation, and time to ICU and hospital discharge).⁸⁹ These

findings are consistent with a systematic review of 16 RCTs and 10 observational studies, which found no difference in delirium outcomes (for example, incidence, duration, or severity of delirium, sedation status, length of stay, cognitive function, or death) in patients treated with haloperidol, second generation antipsychotic drugs, or placebo.⁹⁰ Three multicenter RCTs have found that dexmedetomidine, compared with propofol or benzodiazepines, significantly reduces duration of delirium in mechanically ventilated patients.^{83 84 91} In a placebo controlled trial to assess the effectiveness of dexmedetomidine in treating agitated delirium, delirium resolved a median of 16 (95% confidence interval 3.0 to 28.0) hours earlier ($P=0.01$; van Elteren site adjusted $P=0.04$) and time ventilator-free during the first seven days after randomization was reduced by a median of 17 (4.0 to 33.2) hours compared with placebo ($P=0.01$).⁹²

Acetaminophen

Recently, in an RCT of 120 cardiac patients aged 60 years and older (only 16% were women) following on-pump coronary artery bypass graft (CABG) or CABG with valvular surgery, treatment with postoperative intravenous acetaminophen combined with intravenous propofol or dexmedetomidine had an 18% (95% confidence interval 5% to 32%) lower rate of in-hospital delirium than did placebo combined with propofol or dexmedetomidine.⁹³ By contrast, patients receiving dexmedetomidine rather than propofol had no significant difference in delirium rates (17% v 21%; difference -4%, -18% to 10%; hazard ratio 0.8, 0.4 to 1.9; $P=0.54$).⁹³ As this trial included only CABG patients, further study is needed to examine the effects of intravenous acetaminophen in a general ICU patient population, as well as broader investigation of underlying physiologic

Table 2 | Alternative instruments that have been used to assess delirium in critically ill patients; only 3 of 7 have been validated

Instrument	Features assessed	Time period	Scoring	Assessor	Comparator	No	Results
Delirium Detection Score ⁵⁸	Agitation, anxiety, hallucination, orientation, seizures, tremor, paroxysmal sweating, altered sleep-wake rhythm	Longer moment in time assessment	0-56	Clinical physicians and nurses	Sedation-Agitation Scale and defined clinical assessment	1073	69% sensitivity and 75% specificity if score ≥ 8
Neelon and Champagne Confusion Scale (NEECHAM) ⁵⁹	Attention, command, orientation, appearance, motor, verbal, vital function, oxygen saturation, urinary continence	Short moment in time assessment	0-30	Clinical nurses	Psychiatric intern assessment with DSM-IV criteria	105	97% sensitivity and 83% specificity if score ≥ 24
Nursing Delirium Screening Scale ⁶⁰	Disorientation, inappropriate behavior, inappropriate communication, hallucination, psychomotor retardation	Assessment over nursing shift or moment in time assessment	0-10	Research physicians and nurses	Psychiatric intern assessment with DSM-IV criteria	156	82% sensitivity and 83% specificity if score ≥ 2
Cognitive Test for Delirium ⁶¹	Orientation, attention span, memory, comprehension and conceptual reasoning, vigilance	Longer moment in time assessment	0-30	Psychologist technician	Psychiatric expert assessment with DSM-III-R criteria	103	100% sensitivity and 95% specificity if score $\leq 18^*$
Abbreviated Cognitive Test for Delirium ⁶²	Visual attention span, recognition memory	Short moment in time assessment	0-24	Psychologist technician	Psychiatric expert assessment with DSM-III-R criteria	100	95% sensitivity and 99% specificity if score $\leq 10^*$
Family Confusion Assessment Method (FAM-CAM) ⁶³	Fluctuation, inattention, disorganized thinking or altered consciousness, disorientation, perceptual disturbances, inappropriate behaviors, disorientation, drowsiness in daytime	Assessment by caregiver	Positive or negative	Trained research associate	CAM-ICU-7	17	--†
Sour Seven Questionnaire ⁶³	Altered level of awareness, reduced attentiveness, fluctuation, disordered thinking, disorganized behavior, impaired eating/drinking, difficulty in mobility	Assessment by caregiver	0-18	Trained research associate	CAM	17	--‡

CAM=Confusion Assessment Method; DSM=Diagnostic and Statistical Manual of Mental Disorders; ICU=intensive care unit.

*17 caregivers and critically ill patients; descriptive study (feasibility); not powered to test validity or reliability of tool.

†52 paired dyads of caregivers and elderly patients; CAM as comparator; 88% sensitivity and 98% specificity.⁶⁴

‡80 participants; seniors over age of 65 years; 92% sensitivity and 100% specificity.⁶⁵

mechanisms (for example, immune mediated versus avoidance of hyperthermia).

Valproate

Recently, valproate has been administered to critically ill patients to treat agitation and delirium, but few published reports are available to support this practice.⁹⁴⁻⁹⁷ Valproate has both intravenous and enteral formulations, has a low drug acquisition cost, and has not been associated with respiratory depression, hemodynamic derangements, or delirium. In a recent retrospective cohort study, 53 patients received an average of 1500 mg/day of valproate for agitation started at a median day 7 of an ICU stay and continued for a median of 7 days.⁹⁸ Whereas agitation and delirium were increasing in frequency over time before initiation of valproate, administration was associated with reduced agitation (61% on day 3 of valproate versus 96% on day 1; $P<0.0001$) and delirium (49% v 68%; $P=0.012$).⁹⁸ The use of opioids (95% v 61%; $P=0.02$) and dexmedetomidine (47% v 24%; $P=0.004$) was also reduced. Side effects were common, however, with hyperammonemia reported in 19% and thrombocytopenia in 13% of patients receiving valproate.⁹⁸ Two other retrospective reviews of prescription of valproate for agitation and delirium found similar results. In a retrospective descriptive analysis of a medical-surgical ICU patient population, median starting dose was 6.2 mg/day with median levels reaching 34.9 (interquartile range 22.3-78.1) $\mu\text{g/mL}$; median duration of therapy was 8 days, and a trend was seen toward reduced rates of agitation and delirium (48% v 17% and 85% v 63%, respectively).⁹⁹ In the largest retrospective cohort of

80 patients, 35 of whom received valproate alone and 45 of whom received valproate in combination with antipsychotics, resolution of delirium occurred in 55% of patients (69% in valproic acid alone group and 44% in valproic acid plus antipsychotics group; $P=0.03$).¹⁰⁰ Furthermore, delirium incidence decreased from initiation of valproate to day 3 of therapy from 93% to 68% ($P<0.01$), with no change in agitation (64% v 63%; $P=0.28$). Investigation in placebo controlled randomized trials is needed, however, to determine the value of valproate alone or as an adjunct in the management of ICU delirium.¹⁰¹

Diagnostic tools for delirium

Biomarkers

Biomarkers are an emerging tool that may further our understanding of the pathophysiology of delirium and possibly lead to earlier diagnosis and better risk stratification. Several biomarkers have been associated with delirium during critical illness. The S100B protein is an indicator of glial activation and/or death and thus is a non-specific marker of brain injury. This marker has also been validated as an indicator of blood-brain barrier injury.¹⁰² Plasma S100B concentration has been found to be elevated in patients with delirium.¹⁰³⁻¹⁰⁵ In a population with septic shock in which the incidence of delirium was approximately 50%, the odds ratio for developing delirium was 18 (95% confidence interval 2 to 196) with a S100B protein concentration of 0.15 $\mu\text{g/L}$ or greater. Other research exploring general systemic inflammatory biomarkers as predictors of delirium include a pilot study of patients enrolled in the Maximizing Efficacy of Targeted Sedation

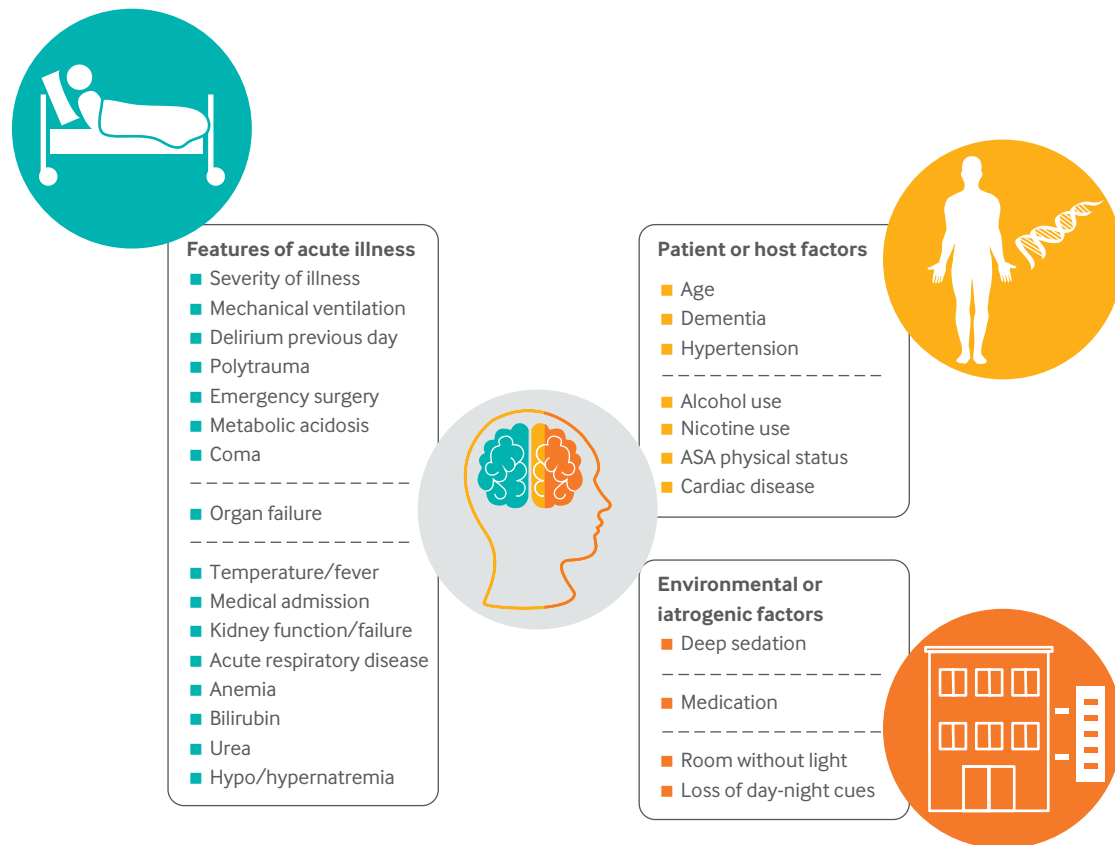


Fig 3 | Predisposing risk factors and precipitating factors for delirium. ASA=American Society of Anesthesiologists

and Reducing Neurological Dysfunction (MENDS) trial evaluating 87 critically ill patients, most of whom had sepsis on admission to the ICU.¹⁰⁶ This study found that higher baseline concentrations of procalcitonin or C reactive protein were associated with more days with delirium.¹⁰⁶

Additional serum biomarkers shown to be elevated in patients with delirium include brain derived neurotrophic factor, neuron specific enolase, interleukins, and cortisol.^{107 108} For example, a retrospective analysis of the large sepsis cohort, the Molecular Epidemiology of Severe Sepsis in the ICU (MESSI) study,¹⁰⁹ examined neuron specific enolase concentrations in the initial blood drawn at presentation to the emergency department for patients directly admitted to the ICU and in blood drawn during or just after decompensation for patients transferred to the ICU from the medical ward. Each doubling of the concentration was associated with a 5.2% (95% confidence interval 3.2% to 7.2%) increase in the risk of delirium; a concentration >12.5 µg/L was independently associated with a 29.3% (8.8% to 49.8%; $P=0.005$) increase in the risk of delirium.¹⁰⁹ Interestingly, in the Bringing to light the Risk factors And Incidence of Neuropsychological dysfunction in ICU survivors (BRAIN-ICU) study, in which 70% of 134 patients experienced delirium after admission for acute respiratory failure and/or shock, those with increased concentrations of plasminogen activator

inhibitor-1, E-selectin, and S100B experienced fewer delirium-free and coma-free days and more delirium days than those without increased concentrations, after adjustment for potential confounders.¹¹⁰ In a nested prospective cohort of 138 patients from the Awakening and Breathing Controlled RCT, most of whom were over 65 years of age, elevated matrix metalloproteinase-9, protein C, and soluble tumor necrosis factor receptor-1 within two days of enrollment increased the probability of transitioning to delirium during ICU stay.¹¹¹ Very few studies have used a time series design. In a cohort of patients at high risk of developing delirium ($n=50$; predicted risk of delirium >40% using the PRE-DELIRIC model), no differences in the concentrations of any biomarkers immediately before or after the onset of delirium were seen.¹¹² As the concentrations of biomarkers and the risk of transitioning to delirium may be fluid over time, serial measurements of biomarkers may further elucidate the interplay between inflammation and neuronal injury.

Neuroimaging

In addition to biomarkers, neuroimaging is an emerging diagnostic tool for delirium. Modalities such as functional magnetic resonance imaging (MRI), diffusion tensor imaging, arterial spin labeling, and positron emission tomography have revealed nonspecific or diffuse changes over time, including cortical atrophy, ventricular enlargement,

and white matter hyperdensities.^{113 114} Research is preliminary, with three studies currently registered on clinicaltrials.gov: two examining delirium as a complication in the postoperative course, using a range of imaging modalities (for example, functional MRI, carbon dioxide stress testing; NCT02126215 and NCT03110185) and one study investigating structural and diffusion tensor imaging sequences at one year after discharge from the ICU and their relation with cognitive performance (NCT03562689).

Electroencephalography

Lastly, an emerging tool for the diagnosis of delirium is electroencephalography. Delirium is associated with slowing of background activity on electroencephalography, specifically an increased relative delta power (1-4 Hz).¹¹⁵⁻¹¹⁸ Electroencephalograph based monitoring and quantification has recently shown potential for detecting delirium in routine daily practice. A single center observational study of 54 patients (26 adults with delirium, age and sex matched with 28 non-delirious

Table 3 | Studies of delirium and long term outcomes in patients with critical illness

Date	Characteristic	Study cohort	Primary outcome	Results
Girard et al, 2010 ¹⁰	Prospective cohort	126 medical ICU patients; mechanically ventilated	Cognitive outcome at 3 and 12 months	Duration of delirium independently associated with long term cognitive outcomes
Gunther et al, 2012 ¹³⁶	Prospective cohort	47 patients from medical and surgical ICUs with respiratory failure or shock	3-Tesla brain MRI at discharge and 3 and 12 months follow-up	Delirium duration associated with white matter disruption at both discharge and 3 months; white matter disruption associated with worse cognitive scores at 12 months
Morandi et al, 2012 ¹¹⁴	Prospective cohort	47 patients from medical and surgical ICUs with respiratory failure or shock	3-Tesla brain MRI at discharge and 3 and 12 months follow-up	Longer duration of delirium associated with smaller brain volumes up to 3 months after discharge; smaller brain volumes associated with long term cognitive impairment up to 12 months
Van den Boogaard et al, 2012 ⁸	Prospective cohort	1292 patients; 272 with delirium and 1020 without delirium	Short Form-36v1, checklist individual strength-fatigue, and cognitive failure questionnaire at 18 months	Intensive care survivors with delirium had similar adjusted health related quality of life; duration of delirium associated with long term cognitive impairment
Pandharipande et al, 2013 ¹¹	Prospective cohort	821 patients with respiratory failure or shock in medical or surgical ICU	Cognitive outcome at 3 and 12 months	Days of delirium in hospital associated with worse global cognition and executive function at 3 and 12 months
Brummel et al, 2014 ¹⁷	Prospective cohort nested within RCT of paired sedation and ventilator weaning strategy	126 survivors of critical illness	Katz activities of daily living, Functional Activities Questionnaire, Medical Outcomes Study 36-item Short Form, and Awareness Questionnaire	Days delirious independently associated with increased odds of disability in activities of daily living and worse motor-sensory function at 1 year
Jackson et al, 2015 ¹³⁹	Prospective cohort	47 survivors of medical, surgical, or cardiac ICUs	Functional MRI at hospital discharge and 3 months post-discharge	Delirium not associated with distinct or abnormal brain activation patterns
Sakuramoto et al, 2015 ¹⁴⁶	Prospective cohort	79 survivors of critical illness	MMSE at the time of hospital discharge	Patients experiencing delirium (63%) in ICU had higher rates of cognitive impairment at time of hospital discharge (28% v 3%; P=0.03); averaged ICDSC score over course of ICU stay positively associated with cognitive impairment (OR 1.6, 95% CI 1.02 to 2.54; P=0.041)
Wolters et al, 2017 ¹⁴⁰	Prospective cohort	567 one year survivors from medical-surgical ICU	Cognitive failures questionnaire at 1 year	Days delirious independently associated with greater self-reported cognitive problems
Girard et al, 2018 ⁵¹	Prospective cohort	586 patients managed in medical or surgical ICU with respiratory failure, septic or cardiogenic shock, or both	Repeatable Battery for the Assessment of Neuropsychological Status, MMSE, and Trails Making Test B at 3 and 12 months	Sedative associated delirium, septic delirium, hypoxic delirium, and unclassified delirium were each associated with long term cognitive impairment, whereas metabolic delirium was not
Bruck et al, 2018 ¹⁴¹	Prospective cohort	125 ICU survivors	Cognitive failures questionnaire at 3 months	Neither severe sepsis nor ICU delirium was associated with self-rated cognitive function 3 months after ICU stay
Mitchell et al, 2018 ¹⁴²	Prospective cohort	148 medical or surgical ICU survivors	Repeatable Battery for the Assessment of Neuropsychological Status and Trails Making Tests at 3 and 6 months	ICU delirium associated with impaired information processing speed and executive function at 6 months follow-up
Schulte, et al, 2019 ¹⁴⁴	Retrospective cohort; Mayo Clinic Study of Aging and ICU	372 of 3673 participants who had ICU admission	Trails Making Test B, Digit Symbol Substitution Test, Boston Naming Test and Category Fluency, Wechsler Memory Scale-Revised Logical Memory II, Wechsler Memory Scale-Revised Visual Reproduction-II, Auditory Verbal Learning Test, Wechsler Adult Intelligence Scale-Revised Picture Completion test, Wechsler Adult Intelligence Scale-Revised Block Design test	ICU admission associated with greater long term cognitive decline compared with patients without ICU admission; findings were more pronounced in those who developed delirium while in ICU
Zhang et al, 2019 ¹⁴³	RCT; low dose dexmedetomidine v placebo	700 elderly patients admitted to non-cardiac ICU	Telephone interview at 3 year follow-up; modified Telephone Interview for Cognitive Status	Low dose dexmedetomidine infusion did not significantly change 3 year overall survival but increased survival up to 2 years and improved cognitive function and quality of life in 3 year survivors
Bulic et al, 2020 ¹⁴⁵	Prospective cohort	103 participants; 36% developed delirium	MMSE, Telephone Interview for Cognitive Status	Delirium in ICU independently associated with short term but not long term cognitive function, and with long term PTSD symptoms

ICDSC=Intensive Care Delirium Screening Checklist; ICU=intensive care unit; MMSE=Mini Mental State Examination; MRI=magnetic resonance imaging; OR=odds ratio; PTSD=post-traumatic stress disorder; RCT=randomized controlled trial.

patients after cardiothoracic surgery), found that electrode derivation of relative delta power in frontal and parietal leads was able to distinguish delirium from a non-delirious state (a cut-off value of 0.37 had a sensitivity of 100% and specificity of 96%).¹¹⁷ More recently, a prospective multicenter study of 159 older (>60 years) patients who had major surgery requiring ICU care for more than 48 hours showed that delirious patients (84/276 assessments) had a 50% increase in relative delta power compared with non-delirious patients (area under the receiver operating characteristic curve of relative delta power was 0.75, 95% confidence interval 0.69 to 0.81).¹¹⁹ Interestingly, delta power correlated with severity of delirium, showing a dose dependence that suggested that understanding slow waves may lead to important insights into the pathology of delirium. Specifically, understanding the biologic inter-relationships of slow waves and connectivity of upper brainstem networks will be critical in understanding the pathogenesis of delirium. Although these data are preliminary and the participants were not sedated, validation studies conducted in unselected ICU patient populations and a rigorous evaluation of residual effects of sedatives would be highly valuable to the field. A prospective multicenter stepped wedge cluster randomized trial in four ICUs in Germany is under way to compare standard screening methods and delta power (DeltaScan) on delirium detection rates and length of stay in the ICU (NCT03735940). Electroencephalography based testing has the potential to be an objective and informative measure if it is able to detect delirium on a continuous scale, without the limits of language or sensory barriers (such as hearing impairment).

Long term cognitive impairment Epidemiology

For many people, an acute hospital admission, especially one that includes an ICU stay, results in new or worsening cognitive decline.^{11 120-124} In a multicenter cohort of 821 critically ill patients with respiratory failure or shock, one in four ICU survivors, across all adult age groups, had cognitive impairment one year after discharge, equating in severity to that of having mild Alzheimer's disease or moderate traumatic brain injury.¹¹ In a recent systematic review of 46 studies, aggregated frequency of cognitive impairment after critical illness was 35% and 45% at three and six months' follow-up, respectively.¹²⁵ Interestingly, survivors of acute respiratory distress syndrome had a higher prevalence of cognitive impairment than did a mixed ICU survivor population (approximately 80% compared with 50% prevalence at three month follow-up).¹²⁵ Cognitive impairment seems to improve on average, however, from three to six months' follow-up,^{125 126} and any existing differences between survivor populations do not persist beyond three months.¹²⁵ Regardless of improvements that might be seen in early recovery, a substantial percentage of patients have persistent long term cognitive impairment, as high as 42% and

46% at one and two years after ICU discharge.^{11 126-129} Post-intensive care syndrome describes the disability that remains in ICU survivorship; it comprises impairment in cognition, psychological health, and physical function in the ICU survivor and the consequent psychological health of family members of the survivor.¹³⁰

Pre-existing cognitive impairment

Although patients can develop de novo cognitive impairment after critical illness,^{131 132} hospital admissions may also adversely affect pre-existing cognitive deficits, perhaps owing to the effects of decreased cognitive reserve and accelerated processes of cognitive impairment. Recently, in a multicenter prospective cohort study of both planned and unplanned admissions (n=2467), the rate of cognitive impairment in a mixed patient population at baseline was determined to be 6% by patient or substitute decision maker completed questionnaire.¹³³ These results are comparable to those of a study using data from the Health and Retirement Study, a nationally representative survey of US residents (1998-2006), in which 6.1% (95% confidence interval 4.2% to 8.0%) of eventual survivors of sepsis (n=1194) had moderate/severe cognitive impairment just before ICU admission, with the prevalence increasing to 16.7% (13.8% to 19.7%) at the first survey after severe sepsis (P<0.001 by χ^2).¹³² Reported incidence of pre-existing cognitive impairment was higher (35%; n=136/387) in patients admitted to or transferred to the ICU in a population based study (Mayo Clinic Study of Aging), than in older patients admitted to hospital who did not need to be admitted to ICU (18%; n=391/1733).¹³⁴ After adjustment for important covariates, mild cognitive impairment remained a significant predictor of ICU admission (hazard ratio 1.50, 1.15 to 1.96).¹³⁴ Clinical factors associated with cognitive impairment included higher age, male sex, previous stroke, and poor self-reported health.¹³⁴ Pre-existing cognitive impairment, in addition to being a predictor of ICU admission and cognitive decline, also seems to influence disability. In a prospective cohort of 754 ICU survivors aged 70 years or older undergoing routine cognitive screening, the presence of pre-existing mild or moderate cognitive impairment predicted increased disability at six months (adjusted relative risks 1.19 (1.04 to 1.36) and 1.16 (1.02 to 1.32), respectively).¹³⁵

Delirium as a risk factor for cognitive impairment

The most consistent risk factor for subsequent long term cognitive impairment identified during critical illness is delirium.^{10 11 51 114 136-146} (table 3). After adjustment for age, education, pre-existing cognitive function, severity of illness, and exposure to sedative drugs, increasing duration of delirium was an independent predictor of worse long term cognitive impairment in a large, multicenter study of survivors of critical illness.¹⁰ In a separate prospective cohort of medical-surgical ICU survivors,

multiple days of delirium in the ICU was associated with increased self-reported cognitive problems at one year follow-up.¹⁴⁰ In another prospective cohort, the average ICDSC score during the ICU stay, a representation of severity of delirium, was positively associated with cognitive impairment at the time of hospital discharge (odds ratio 1.6, 1.02 to 2.54; $P=0.041$).¹⁴⁶

Few studies have examined the pathophysiology of cognitive impairment after critical illness, but the link between delirium and long term cognitive impairment is hypothesized to be mediated directly or indirectly through a systemic inflammatory response inducing the activation of brain parenchymal cells and expression of pro-inflammatory cytokines and inflammatory mediators within the central nervous system.^{147 148} This acute inflammatory response to critical illness may prime microglia, activating them from a resting state. Activated microglia may then perpetuate a state of chronic neuroinflammation and neurotoxicity that, in part, explains long term cognitive impairment.¹⁴⁹ Certain phenotypes of delirium, specifically hypoxic and septic delirium, have been associated with greater risk of long term cognitive impairment.⁵¹ Longer duration of sedative associated delirium was also predictive of worse Repeatable Battery for Assessment of Neuropsychological Status (RBANS) global score at 12 months' follow-up.^{10 11 114 136 140} Interestingly, elevated C reactive protein concentrations, a general marker of inflammation, did not modify the relation between days of delirium and long term cognition.¹⁴⁰ Whether sedative use is associated with the inflammatory response is unclear; a pilot trial is under way to investigate the effect of dexmedetomidine on brain derived neurotrophic factor, neuron specific enolase, and S100B in polytrauma.¹⁵⁰ Most recently, in a secondary analysis of two prospective multicenter cohorts ($n=548$; 88% received mechanical ventilation), neither inflammatory mediators nor coagulation parameters were associated with long term cognitive performance at three or 12 months' follow-up.¹⁵¹

Effect of sedation

The direct relation between sedative agents and cognitive outcome in survivors of critical illness has not been well studied. In a meta-analysis of 10 RCTs and one retrospective cohort study ($n=1216$), the use of lighter sedation was associated with a reduced risk of long term cognitive impairment (odds ratio 0.75, 0.63 to 0.90).¹⁵² Only two of the included studies considered the specific hypothesis that higher doses of sedative and/or analgesic agents may be associated with cognitive impairment after hospital discharge.^{11 129} At 12 months' follow-up, the intervention (received lower benzodiazepine exposure but higher propofol exposure) and usual care groups from the Awakening and Breathing Controlled Trial showed similar cognitive outcomes.¹²⁹ Comparable results were found in a multicenter prospective cohort in which delirium was an independent factor for worse

cognitive function, but no significant association was found between sedative and analgesic medication and long term cognitive impairment.¹¹ Nevertheless, critically ill patients who develop multiple days of delirium in the setting of sedation are at higher risk for long term cognitive impairment than those who are delirious for a shorter duration (or not at all) while sedated,⁵¹ suggesting a complex interaction between sedation, host susceptibility, and long term cognitive outcomes. Thus, RCTs are needed to determine whether interventions that limit sedative exposure and thereby reduce delirium mitigate long term cognitive impairment in survivors of critical illness (fig 4).

Predicting long term cognitive impairment

Studies have so far been unable to identify patients at higher risk of long term cognitive impairment after critical illness using brief cognitive screening tools in the hospital. For example, in a prospective cohort study, performance on neither the Mini Mental State Examination nor MiniCog at the time of hospital discharge predicted cognitive impairment at six months' follow-up.¹⁵⁴ Performance on more sensitive tests of cognitive impairment may have predictive value, but these have not been used in prediction focused investigations to date. A recent systematic review highlighted the absence of prediction models for cognitive impairment after surviving critical illness, whereas two studies have predicted physical function and one study predicted mental health disorders.¹⁵⁵ The lack of predictive ability restricts the ability of clinicians and researchers to adequately risk stratify patients to their individual rehabilitation needs.

Therapies to improve cognitive recovery

A limited number of studies have examined interventions designed to hasten recovery and/or prevent decrements in cognitive function after critical illness.¹⁵⁶ Most studies have focused on intervening during ICU admission to reduce cognitive impairment. Treatments have included strategies for optimizing enteral nutrition, fluids, sedation, weaning, mobilization, cognitive activities, statins, and sleep quality.^{84 129 138 157-160} The only outpatient intervention examined to date was examined in a pilot study of 21 medical and surgical ICU survivors. The Returning to Everyday Tasks Using Rehabilitation Networks (RETURN) study randomized people to either 12 weeks of in-home combined cognitive and physical rehabilitation or usual care (characterized by sporadic rehabilitation).¹⁶¹ Despite nearly equivalent scores on a measure of executive functioning at baseline, patients in the intervention group showed significantly better executive functioning at the end of the intervention and reported fewer disabilities in instrumental activities of daily living.¹⁶¹ Large scale, well designed trials with specific aims and focused research collaboratives with the ability to combine clinical, basic science, and epidemiologic approaches are needed (fig 5).

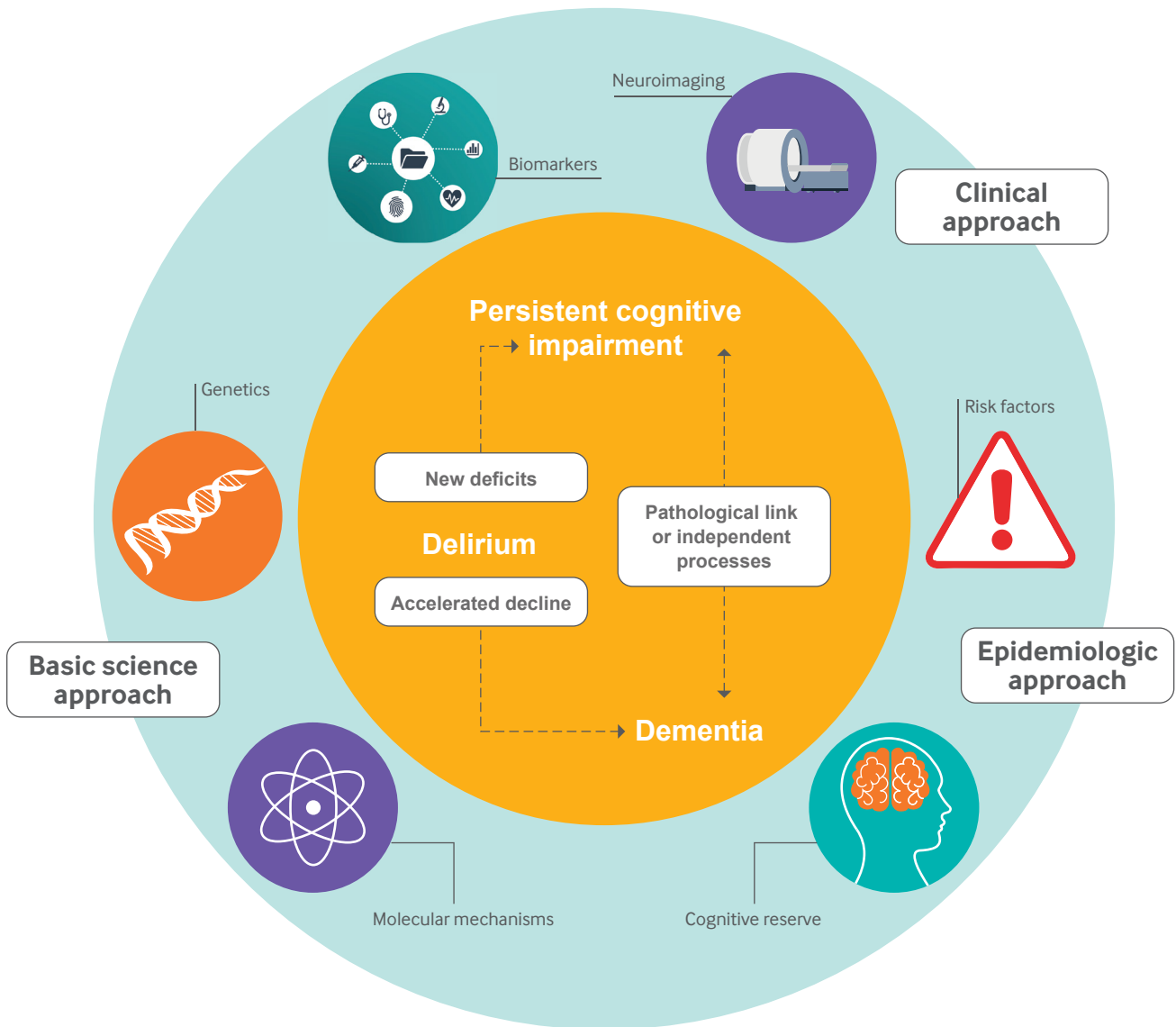


Fig 4 | Conceptual framework for exploring inter-relation between delirium and long term cognitive impairment and between delirium and acceleration of dementia. Source: Inouye and Ferrucci. *J Gerontol A Biol Sci Med Sci* 200⁶¹⁵³

Ongoing trials

A search of clinicaltrials.gov and controlled-trials.com uncovered only two ongoing trials. One is a feasibility trial studying cognitive benefits of prevention and rehabilitation (clinicaltrials.gov NCT03972384). This US based feasibility study is investigating the recruitment, retention, and engagement of a program, Post-Intensive Care Unit Problem Solving (PIC-UPS), during which participants will work for 10 sessions on skills they select as treatment goals. The first session consists of an assessment and selection of goals to be met. In the subsequent sessions, held over 12 weeks and conducted at the participant's home, skills are practiced to reach those goals, assisted by the study team who advise on learning and performing strategies. Some examples of tasks that have been included are gardening, yoga breathing, handwriting, managing

family finances, managing medication, or starting an exercise regimen. Participants in the control group will complete several surveys on enrollment and randomization. Follow-up data collection will be conducted in the home for all participants by a blinded data collector three months after enrollment. The primary outcome will be cognitive performance at three months after discharge from hospital. Secondly, the Improving Recovery and Outcomes Every Day after the ICU (IMPROVE) study (clinicaltrials.gov NCT03095417), is a four arm RCT comprising cognitive training and physical exercise, cognitive control and physical exercise, cognitive training and physical exercise control, and cognitive and physical exercise control.¹⁶² Cognitive training facilitated by online modules in patients' homes and physical exercise will also be administered using internet enabled videoconferencing in small groups.

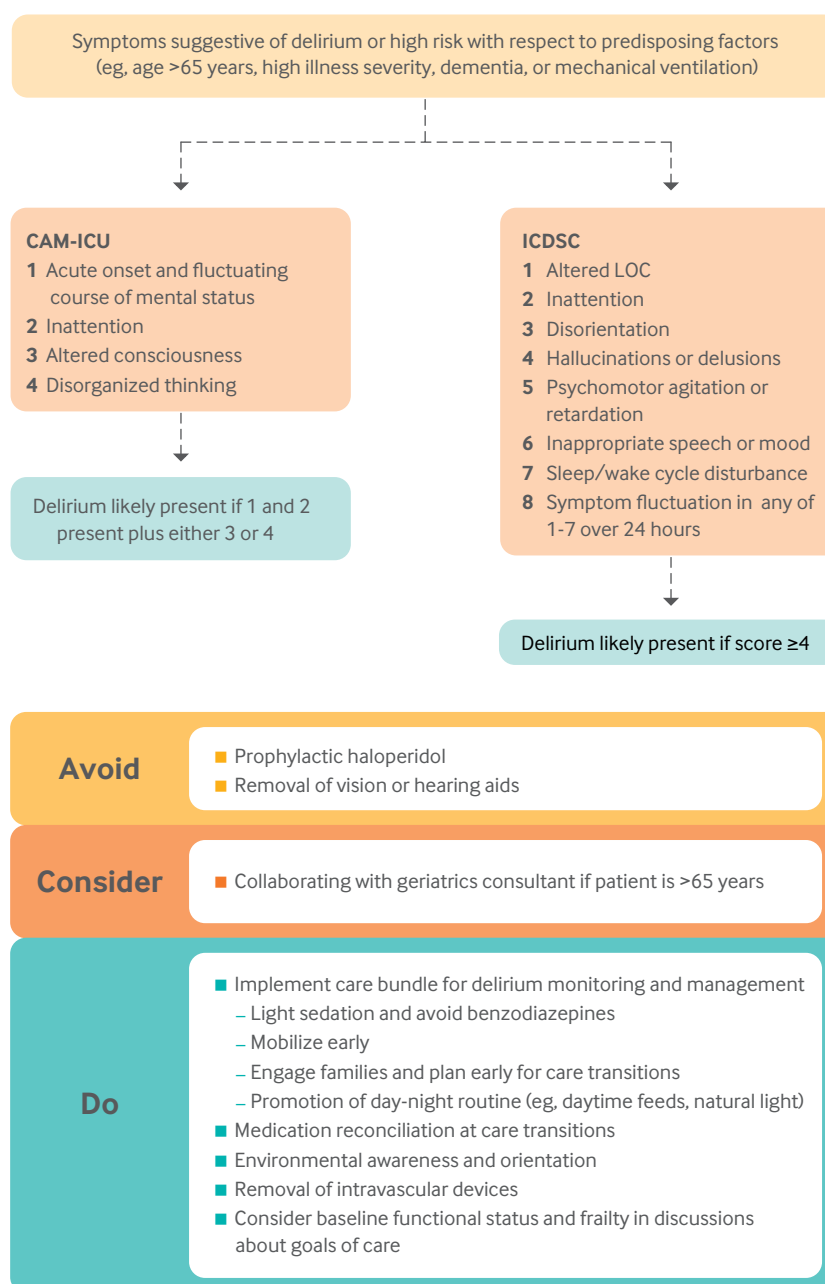


Fig 5 | Populations at high risk, validated screening tools, and prevention/treatment strategies for delirium. CAM-ICU=Confusion Assessment Method for the ICU; ICDSC=Intensive Care Delirium Screening Checklist; LOC=level of consciousness

The primary outcome is cognitive impairment at six months compared with baseline and three months after discharge from ICU.

Guidelines

International guidelines exist for the prevention and management of delirium (table 4). We report the similarities and differences between six international guidelines from the US, UK, Denmark, Germany, Japan, and Panamerica-Iberia: Society of Critical Care Medicine (SCCM),²⁰ National Institute for Health and Care Excellence and Scottish Intercollegiate Guidelines Network (NICE/SIGN),¹⁹ Danish Society of

Anesthesiology and Intensive Care Medicine and the Danish Society of Intensive Care Medicine (DASAIM/DSIT),¹⁶³ German Society of Anaesthesiology and Intensive Care Medicine and German Interdisciplinary Association for Intensive Care and Emergency Medicine (GSAICM/GIAICEM),¹⁶⁴ Japanese Society of Intensive Care Medicine (JSICM),¹⁶⁵ and Panamerican-Iberian Federation of Societies of Critical and Intensive Care Medicine (FEPIMCTI).¹⁸ If multiple iterations for a specific society exist, we chose the most current to discuss. A recent systematic review formally appraised eight clinical practice guidelines by using the AGREE II methodology,¹⁶⁶ including

two versions of each of the GSAICM/GIAICEM, SCCM PADIS (formally PAD), and FEPIMCTI guidelines; the JSICM guidelines were not included. Overall, five of the eight guidelines exceeded the overall quality threshold of 60% in four of six domains.^{21 166}

A total of 73 recommendations have been made with respect to assessment and management of delirium.^{18 19 20 163-165} All guidelines reported on risk factors for delirium; only the SCCM PADIS guidelines comment on transfusion and benzodiazepine use being modifiable. The SCCM PADIS, DASAIM/DSIT, GSAICM/GIAICEM, and JSICM guidelines recommend regular assessment with a validated delirium screening tool, whereas the NICE/SIGN guideline specifically recommends the CAM-ICU or ICDSC. The FEPIMCTI guideline recommends the CAM-ICU, with a caution about its risk of false negatives. With regard to drug treatment, most guidelines do not recommend any agent for the prevention of delirium but recommended minimizing the amount of sedation. All guidelines recommended non-drug multicomponent interventions as a delirium management strategy.

Implementation challenges

Many studies have reported improved patient centered outcomes with implementation of evidence based bundles aimed at reducing ICU acquired

delirium. For example, the use of the Assess, prevent, and manage pain, Both spontaneous awakening trials and spontaneous breathing trials, Choice of analgesia or sedation, Delirium: assess, prevent, and manage, Early mobility and exercise, and Family engagement and empowerment (ABCDEF) bundle in before-after and observational studies has been associated with reduced rates of delirium, shorter duration of mechanical ventilation, and improved survival.^{23 167-172} A prospective cohort study of 1855 patients admitted to two medical ICUs showed that implementation of bundle components reduced duration of mechanical ventilation, length of stay in ICU and hospital, and total ICU and hospital costs.¹⁷³ Also, in this study, duration of mechanical ventilation and length of stay in ICU and hospital were reduced, as was use of midazolam infusion, with implementation of PAD guidelines.¹⁷⁴

Many multifaceted implementation programs have involved the implementation of processes of care and clinical outcomes targeted at screening for delirium, limited use of benzodiazepines, light sedation targets, and early mobilization.^{22 24 168 175} As an example, one program resulted in a reduction in both mean duration of delirium (from 5.6 to 3.3 days; difference -2.2 days; $P<0.001$) and coma days (risk ratio 0.5; $P<0.001$).¹⁷⁵ This program was informed by a previous study by the same group on attitudes,

Table 4 | Summary of international guidelines for delirium in intensive care unit (ICU) patients

Guideline	Organization, country	Title	Scope and purpose	Stakeholder involvement	Rigor of development	Clarity and presentation	Applicability	Editorial independence
SIGN, 2019 ¹⁹	Healthcare Improvement Scotland and Scottish Intercollegiate Guidelines Network, UK	Risk reduction and management of delirium (SIGN 157)	+	+	+	+	+	+
Devlin et al, 2018 ²⁰	Society of Critical Care Medicine, USA	Clinical Practice Guideline for the Prevention and Management of Pain, Agitation/Sedation, Delirium, Immobility, Sleep Disruption in Adult Patients in the ICU	+	+	+	+	+/-	+
Fonsmark et al, 2015 ¹⁶³	Danish Society of Anesthesiology and Intensive Care Medicine (DASAIM) and Danish Society of Intensive Care Medicine (DSIT), Denmark	Danish national sedation strategy. Targeted therapy of discomfort associated with critical illness	+	+/- (no patient engagement)	-	+	-	-
Baron et al, 2015 ¹⁶⁴	German Society of Anaesthesiology and Intensive Care Medicine and German Interdisciplinary Association for Intensive Care and Emergency Medicine, Germany	Evidence and consensus-based guideline for the management of delirium, analgesia, and sedation in intensive care medicine. Revision 2015 (DAS-Guideline 2015)	+	+/- (no patient engagement)	+	+	+/-	+
Committee for the development of Japanese guidelines for the management of Pain, Agitation, and Delirium in intensive care unit, 2014 ¹⁶⁵	Japanese Society of Intensive Care Medicine, Japan	Japanese guidelines for the management of Pain, Agitation, and Delirium in intensive care unit (J-PAD)	+	+/- (no patient engagement)	+	+	+/-	+
Celis-Rodriguez et al, 2020 ¹⁸	Panamerican and Iberian Federation of Societies of Critical and Intensive Care Medicine	Clinical practice guidelines for evidence-based management of sedoanalgesia in critically ill adult patients	+	+/- (no patient engagement)	+	+	+/-	+

knowledge, and practice concerns about bundle implementation.¹⁷⁶ Important barriers identified before the intervention included knowledge deficit, belief that delirium is not preventable, low satisfaction with physician described delirium management, poor collaboration between nursing staff and physicians, and perception that guideline implementation is cumbersome.¹⁷⁶ Given such significant barriers, widespread deployment of such programs could prove difficult, despite the magnitude of potential benefit. More work is needed to better delineate the most effective formats for messaging to frontline workers.

Several publications have reported significant care gaps with regard to guideline implementation, with routine assessment of delirium being the most consistently reported shortfall. For example, over 1101 patient days, a quality improvement initiative found that 39% of delirium assessments were performed compared with 78% agitation assessments.¹⁷⁷ Similarly, in a single center, an audit of practices showed that delirium assessments were performed approximately 30% of the time, with a greater number of assessments being performed during day shifts.¹⁷⁸ A multisite initiative of 69 adult and eight pediatric ICUs reported on

challenges in incorporating the ABCDEF bundle into everyday practice as part of a 20 month nationwide multicenter quality improvement intervention.¹⁶⁹ Routine collection of detailed implementation data is likely essential for program evaluation and generalizability.¹⁷⁹ In a systematic review and meta-analysis, 12 multilevel causal factors affecting implementation outcomes were categorized and subdivided into patient, provider, and organizational factors.¹⁷⁹ Interestingly, organizational factors were rarely reported (for example, hospital type, daily screening frequency, and assessment method used).¹⁷⁹ Organizational factors such as educational strategies, staff knowledge, motivation, and screening compliance may influence the reported incidence of delirium. A standardized framework for reporting such factors would facilitate further insights into different processes of care and their subsequent impact.

Conclusions

Delirium is the most common manifestation of acute brain dysfunction in critically ill patients. It is associated with both poor short term outcomes and adverse long term sequelae related to critical illness. Current strategies for prevention, diagnosis, and treatment are subject to ongoing investigation. In addition to appreciating the acute process of delirium, it is important to recognize that the syndrome, together with other physiologic insults sustained in the ICU, is associated with long term cognitive impairment. The pathophysiology underlying this relation remains a field of active investigation, and uncovering such mechanisms may lead to targeted interventions that improve cognition and quality of life. Advancement of the field will depend on focused research collaboratives with the ability to combine clinical, basic science, and epidemiologic approaches.

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- 1 Vasilevskis EE, Morandi A, Boehm L, et al. Delirium and sedation recognition using validated instruments: reliability of bedside intensive care unit nursing assessments from 2007 to 2010. *J Am Geriatr Soc* 2011;59(Suppl 2):S249-55. doi:10.1111/j.1532-5415.2011.03673.x
- 2 van den Boogaard M, Schoonhoven L, van der Hoeven JG, van Achterberg T, Pickkers P. Incidence and short-term consequences of delirium in critically ill patients: A prospective observational cohort study. *Int J Nurs Stud* 2012;49:775-83. doi:10.1016/j.ijnurstu.2011.11.016
- 3 Girard TD, Pandharipande PP, Ely EW. Delirium in the intensive care unit. *Crit Care* 2008;12(Suppl 3):S3. doi:10.1186/cc6149
- 4 Ely EW, Gautam S, Margolin R, et al. The impact of delirium in the intensive care unit on hospital length of stay. *Intensive Care Med* 2001;27:1892-900. doi:10.1007/s00134-001-1132-2
- 5 Pisani MA, Murphy TE, Van Ness PH, Araujo KL, Inouye SK. Characteristics associated with delirium in older patients in a medical intensive care unit. *Arch Intern Med* 2007;167:1629-34. doi:10.1001/archinte.167.15.1629

RESEARCH QUESTIONS

- Can tools to identify and risk stratify delirium in critically ill patients (eg, electroencephalography) be developed and validated?
 - Can objective tools to measure delirium on a continuous scale, not limited by language or sensory barriers, be developed?
- How can we better understand the attributable risk of delirium in terms of intensive care unit related outcomes?
 - What is the pathophysiology of delirium, and what is its mechanistic relation with long term cognitive impairment?
 - How can we integrate biomarkers of delirium into predictive models of long term cognitive impairment?
 - What is the bidirectional relation between sleep/circadian rhythm and delirium?
- What are the effects of sleep optimization, physical rehabilitation, and cognitive training on delirium and long term cognitive impairment?
 - Can valid intermediate endpoints for testing interventions that may decrease the risk of long term cognitive impairment be identified?
 - Can phenotype driven trials of drug therapies for delirium be designed?
 - Can hybrid trials be conducted to ensure implementation of best practices?
- What is the impact of social networks and family engagement on delirium and long term cognitive impairment?
 - How can socioeconomic inequalities be reduced to improve long term cognitive follow-up and outcomes?
 - How does socially isolated hospital admission during covid-19 affect the risk of delirium and the trajectory of cognitive recovery after critical illness?

PATIENT INVOLVEMENT

A patient and his caregiver reviewed the manuscript and provided useful comments about its content, the presentation of the article, and the use of language. The patient is a middle aged man who experienced delirium during a prolonged stay in intensive care and who remains on medical leave from his job. His wife, who is his primary caregiver, witnessed first-hand her husband's delirium in the intensive care unit, an experience that was very vivid and frightening for her.

- 6 Tasker RC, Menon DK. Critical Care and the Brain. *JAMA* 2016;315:749-50. doi:10.1001/jama.2016.0701
- 7 Wilson JE, Mart MF, Cunningham C, et al. Delirium. *Nat Rev Dis Primers* 2020;6:90. doi:10.1038/s41572-020-00223-4
- 8 van den Boogaard M, Schoonhoven L, Evers AW, van der Hoeven JG, van Achterberg T, Pickkers P. Delirium in critically ill patients: impact on long-term health-related quality of life and cognitive functioning. *Crit Care Med* 2012;40:112-8. doi:10.1097/CCM.0b013e31822e9fc9
- 9 Pandharipande PP, Girard TD, Ely EW. Long-term cognitive impairment after critical illness. *N Engl J Med* 2014;370:185-6.
- 10 Girard TD, Jackson JC, Pandharipande PP, et al. Delirium as a predictor of long-term cognitive impairment in survivors of critical illness. *Crit Care Med* 2010;38:1513-20. doi:10.1097/CCM.0b013e3181e47be1
- 11 Pandharipande PP, Girard TD, Jackson JC, et al. BRAIN-ICU Study Investigators. Long-term cognitive impairment after critical illness. *N Engl J Med* 2013;369:1306-16. doi:10.1056/NEJMoa1301372
- 12 Goldberg TE, Chen C, Wang Y, et al. Association of Delirium With Long-term Cognitive Decline: A Meta-analysis. *JAMA Neurol* 2020;77:1373-81. doi:10.1001/jamaneurol.2020.2273
- 13 Hsieh TT, Saczynski J, Gou RY, et al. SAGES Study Group. Trajectory of Functional Recovery After Postoperative Delirium in Elective Surgery. *Ann Surg* 2017;265:647-53. doi:10.1097/SLA.0000000000001952
- 14 Hsieh SJ, Soto GJ, Hope AA, Ponea A, Gong MN. The association between acute respiratory distress syndrome, delirium, and in-hospital mortality in intensive care unit patients. *Am J Respir Crit Care Med* 2015;191:71-8. doi:10.1164/rccm.201409-16900C
- 15 Zhang Z, Pan L, Ni H. Impact of delirium on clinical outcome in critically ill patients: a meta-analysis. *Gen Hosp Psychiatry* 2013;35:105-11. doi:10.1016/j.genhosppsych.2012.11.003
- 16 Morandi A, Davis D, Fick DM, et al. Delirium superimposed on dementia strongly predicts worse outcomes in older rehabilitation inpatients. *J Am Med Dir Assoc* 2014;15:349-54. doi:10.1016/j.jamda.2013.12.084
- 17 Brummel NE, Jackson JC, Pandharipande PP, et al. Delirium in the ICU and subsequent long-term disability among survivors of mechanical ventilation. *Crit Care Med* 2014;42:369-77. doi:10.1097/CCM.0b013e3182a645bd
- 18 Celis-Rodríguez E, Díaz Cortés JC, Cárdenas Bolívar YR, et al. Evidence-based clinical practice guidelines for the management of sedoanalgesia and delirium in critically ill adult patients. *Med Intensiva* 2020;44:171-84. doi:10.1016/j.medine.2019.07.002
- 19 Scottish Intercollegiate Guidelines Network. SIGN157: Risk reduction and management of delirium. 2019. <https://www.sign.ac.uk/our-guidelines/risk-reduction-and-management-of-delirium/>
- 20 Devlin JW, Skrobik Y, Gélinas C, et al. Clinical Practice Guidelines for the Prevention and Management of Pain, Agitation/Sedation, Delirium, Immobility, and Sleep Disruption in Adult Patients in the ICU. *Crit Care Med* 2018;46:e825-73. doi:10.1097/CCM.00000000000003299
- 21 Tsuruta R, Fujita M. Comparison of clinical practice guidelines for the management of pain, agitation, and delirium in critically ill adult patients. *Acute Med Surg* 2018;5:207-12. doi:10.1002/ams2.337
- 22 Pun BT, Balas MC, Barnes-Daly MA, et al. Caring for Critically Ill Patients with the ABCDEF Bundle: Results of the ICU Liberation Collaborative in Over 15,000 Adults. *Crit Care Med* 2019;47:3-14. doi:10.1097/CCM.00000000000003482
- 23 Ely EW. The ABCDEF Bundle: Science and Philosophy of How ICU Liberation Serves Patients and Families. *Crit Care Med* 2017;45:321-30. doi:10.1097/CCM.00000000000002175
- 24 Barnes-Daly MA, Phillips G, Ely EW. Improving Hospital Survival and Reducing Brain Dysfunction at Seven California Community Hospitals: Implementing PAD Guidelines Via the ABCDEF Bundle in 6,064 Patients. *Crit Care Med* 2017;45:171-8. doi:10.1097/CCM.00000000000002149
- 25 Burry LD, Hutton B, Guenette M, et al. Comparison of pharmacological and non-pharmacological interventions to prevent delirium in critically ill patients: a protocol for a systematic review incorporating network meta-analyses. *Syst Rev* 2016;5:153. doi:10.1186/s13643-016-0327-0
- 26 O'Hanlon S, Inouye SK. Delirium: a missing piece in the COVID-19 pandemic puzzle. *Age Ageing* 2020;49:497-8. doi:10.1093/ageing/afaa094
- 27 Helms J, Kremer S, Merdji H, et al. Delirium and encephalopathy in severe COVID-19: a cohort analysis of ICU patients. *Crit Care* 2020;24:491. doi:10.1186/s13054-020-03200-1
- 28 D'Ardes D, Carrarini C, Russo M, et al. Low molecular weight heparin in COVID-19 patients prevents delirium and shortens hospitalization. *Neurol Sci* 2021;42:1527-30. doi:10.1007/s10072-020-04887-4
- 29 Mummery RS, Rider CC. Characterization of the heparin-binding properties of IL-6. *J Immunol* 2000;165:5671-9. doi:10.4049/jimmunol.165.10.5671
- 30 Zhou F, Yu T, Du R, et al. Clinical course and risk factors for mortality of adult inpatients with COVID-19 in Wuhan, China: a retrospective cohort study. *Lancet* 2020;395:1054-62. doi:10.1016/S0140-6736(20)30566-3
- 31 Liu K, Nakamura K, Katsukawa H, et al. ABCDEF Bundle and Supportive ICU Practices for Patients With Coronavirus Disease 2019 Infection: An International Point Prevalence Study. *Crit Care Explor* 2021;3:e0353. doi:10.1097/CCE.0000000000000353
- 32 Aziz S, Arabi YM, Alhazzani W, et al. Managing ICU surge during the COVID-19 crisis: rapid guidelines. *Intensive Care Med* 2020;46:1303-25. doi:10.1007/s00134-020-06092-5
- 33 Woolf SH, Chapman DA, Sabo RT, Weinberger DM, Hill L. Excess Deaths From COVID-19 and Other Causes, March-April 2020. *JAMA* 2020;324:510-3. doi:10.1001/jama.2020.11787
- 34 Pun BT, Badenes R, Heras La Calle G, et al. COVID-19 Intensive Care International Study Group. Prevalence and risk factors for delirium in critically ill patients with COVID-19 (COVID-D): a multicentre cohort study. *Lancet Respir Med* 2021;9:239-50. doi:10.1016/S2213-2600(20)30552-X
- 35 Ahmed H, Patel K, Greenwood DC, et al. Long-term clinical outcomes in survivors of severe acute respiratory syndrome and Middle East respiratory syndrome coronavirus outbreaks after hospitalisation or ICU admission: A systematic review and meta-analysis. *J Rehabil Med* 2020;52:jrm00063.
- 36 Carfi A, Bernabei R, Landi F, Gemelli Against COVID-19 Post-Acute Care Study Group. Persistent Symptoms in Patients After Acute COVID-19. *JAMA* 2020;324:603-5. doi:10.1001/jama.2020.12603
- 37 Garrigues E, Janvier P, Kherabi Y, et al. Post-discharge persistent symptoms and health-related quality of life after hospitalization for COVID-19. *J Infect* 2020;81:e4-6. doi:10.1016/j.jinf.2020.08.029
- 38 Ramani C, Davis EM, Kim JS, Provencio JJ, Enfield KB, Kadl A. Post-ICU COVID-19 Outcomes: A Case Series. *Chest* 2021;159:215-8. doi:10.1016/j.chest.2020.08.2056
- 39 Tay MRJ, Low YH, Lim CCT, et al. Covert Subclinical Neurocognitive Sequelae During the Rehabilitation Course of Severe Coronavirus Disease 2019. *Am J Phys Med Rehabil* 2021;100:39-43. doi:10.1097/PHM.0000000000001633
- 40 Chopra V, Flanders SA, O'Malley M, Malani AN, Prescott HC. Sixty-Day Outcomes Among Patients Hospitalized With COVID-19. *Ann Intern Med* 2021;174:576-8. doi:10.7326/M20-5661
- 41 Francis J, Kapoor WN. Delirium in hospitalized elderly. *J Gen Intern Med* 1990;5:65-79. doi:10.1007/BF02602312
- 42 Liptzin B, Levkoff SE. An empirical study of delirium subtypes. *Br J Psychiatry* 1992;161:843-5. doi:10.1192/bjp.161.6.843
- 43 Meagher DJ, Trezpac PT. Motoric subtypes of delirium. *Semin Clin Neuropsychiatry* 2000;5:75-85.
- 44 Peterson JF, Pun BT, Dittus RS, et al. Delirium and its motoric subtypes: a study of 614 critically ill patients. *J Am Geriatr Soc* 2006;54:479-84. doi:10.1111/j.1532-5415.2005.00621.x
- 45 Salluh JJ, Soares M, Teles JM, et al. Delirium Epidemiology in Critical Care Study Group. Delirium epidemiology in critical care (DECCA): an international study. *Crit Care* 2010;14:R210. doi:10.1186/cc9333
- 46 Agarwal V, O'Neill PJ, Cotton BA, et al. Prevalence and risk factors for development of delirium in burn intensive care unit patients. *J Burn Care Res* 2010;31:706-15. doi:10.1097/BCR.0b013e3181eebee9
- 47 Roberts B, Rickard CM, Rajbhandari D, et al. Multicentre study of delirium in ICU patients using a simple screening tool. *Aust Crit Care* 2005;18:6, 8-9, 11-4 passim. doi:10.1016/S1036-7314(05)80019-0
- 48 Ely EW, Margolin R, Francis J, et al. Evaluation of delirium in critically ill patients: validation of the Confusion Assessment Method for the Intensive Care Unit (CAM-ICU). *Crit Care Med* 2001;29:1370-9. doi:10.1097/00003246-200107000-00012
- 49 Van Rompaey B, Schuurmans MJ, Shortridge-Baggett LM, Truijens S, Elseviers M, Bossaert L. A comparison of the CAM-ICU and the NEECHAM Confusion Scale in intensive care delirium assessment: an observational study in non-intubated patients. *Crit Care* 2008;12:R16. doi:10.1186/cc6790
- 50 Patel SB, Poston JT, Pohlman A, Hall JB, Kress JP. Rapidly reversible, sedation-related delirium versus persistent delirium in the intensive care unit. *Am J Respir Crit Care Med* 2014;189:658-65. doi:10.1164/rccm.201310-1815OC
- 51 Girard TD, Thompson JL, Pandharipande PP, et al. Clinical phenotypes of delirium during critical illness and severity of subsequent long-term cognitive impairment: a prospective cohort study. *Lancet Respir Med* 2018;6:213-22. doi:10.1016/S2213-2600(18)30062-6
- 52 Salluh JJ, Wang H, Schneider EB, et al. Outcome of delirium in critically ill patients: systematic review and meta-analysis. *BMJ* 2015;350:h2538. doi:10.1136/bmj.h2538
- 53 Ely EW, Inouye SK, Bernard GR, et al. Delirium in mechanically ventilated patients: validity and reliability of the confusion assessment method for the intensive care unit (CAM-ICU). *JAMA* 2001;286:2703-10. doi:10.1001/jama.286.21.2703
- 54 Pisani MA, Kong SY, Kasl SV, Murphy TE, Araujo KL, Van Ness PH. Days of delirium are associated with 1-year mortality in an older intensive

- care unit population. *Am J Respir Crit Care Med* 2009;180:1092-7. doi:10.1164/rccm.200904-0537OC
- 55 Wolters AE, van Dijk D, Pasma W, et al. Long-term outcome of delirium during intensive care unit stay in survivors of critical illness: a prospective cohort study. *Crit Care* 2014;18:R125. doi:10.1186/cc13929
 - 56 Klein Klouwenberg PM, Zaal IJ, Spitoni C, et al. The attributable mortality of delirium in critically ill patients: prospective cohort study. *BMJ* 2014;349:g6652. doi:10.1136/bmj.g6652
 - 57 Vasilevskis EE, Chandrasekhar R, Holtz CH, et al. The Cost of ICU Delirium and Coma in the Intensive Care Unit Patient. *Med Care* 2018;56:890-7. doi:10.1097/MLR.0000000000000975
 - 58 Otter H, Martin J, Bäsel K, et al. Validity and reliability of the DDS for severity of delirium in the ICU. *Neurocrit Care* 2005;2:150-8. doi:10.1385/NCC.2.2:150
 - 59 Immers HE, Schuurmans MJ, van de Bijl JJ. Recognition of delirium in ICU patients: a diagnostic study of the NEECHAM confusion scale in ICU patients. *BMC Nurs* 2005;4:7. doi:10.1186/1472-6955-4-7
 - 60 Neufeld KJ, Hayat MJ, Coughlin JM, et al. Evaluation of two intensive care delirium screening tools for non-critically ill hospitalized patients. *Psychosomatics* 2011;52:133-40. doi:10.1016/j.psych.2010.12.018
 - 61 Hart RP, Levenson JL, Sessler CN, Best AM, Schwartz SM, Rutherford LE. Validation of a cognitive test for delirium in medical ICU patients. *Psychosomatics* 1996;37:533-46. doi:10.1016/S0033-3182(96)71517-7
 - 62 Hart RP, Best AM, Sessler CN, Levenson JL. Abbreviated cognitive test for delirium. *J Psychosom Res* 1997;43:417-23. doi:10.1016/S0022-3999(97)00140-2
 - 63 Krewulak KD, Sept BG, Stelfox HT, et al. Feasibility and acceptability of family administration of delirium detection tools in the intensive care unit: a patient-oriented pilot study. *CMAJ Open* 2019;7:E294-9. doi:10.9778/cmajo.20180123
 - 64 Steis MR, Evans L, Hirschman KB, et al. Screening for delirium using family caregivers: convergent validity of the Family Confusion Assessment Method and interviewer-rated Confusion Assessment Method. *J Am Geriatr Soc* 2012;60:2121-6. doi:10.1111/j.1532-5415.2012.04200.x
 - 65 Shulman RW, Kalra S, Jiang JZ. Validation of the Sour Seven Questionnaire for screening delirium in hospitalized seniors by informal caregivers and untrained nurses. *BMC Geriatr* 2016;16:44. doi:10.1186/s12877-016-0217-2
 - 66 Bergeron N, Dubois MJ, Dumont M, Dial S, Skrobik Y. Intensive Care Delirium Screening Checklist: evaluation of a new screening tool. *Intensive Care Med* 2001;27:859-64. doi:10.1007/s001340100909
 - 67 Zaal IJ, Devlin JW, Peelen LM, Slooter AJ. A systematic review of risk factors for delirium in the ICU. *Crit Care Med* 2015;43:40-7. doi:10.1097/CCM.0000000000000625
 - 68 Hipp DM, Ely EW. Pharmacological and nonpharmacological management of delirium in critically ill patients. *Neurotherapeutics* 2012;9:158-75. doi:10.1007/s13311-011-0102-9
 - 69 Pandharipande P, Cotton BA, Shintani A, et al. Prevalence and risk factors for development of delirium in surgical and trauma intensive care unit patients. *J Trauma* 2008;65:34-41. doi:10.1097/TA.0b013e31814b2c4d
 - 70 Burry LD, Williamson DR, Mehta S, et al. Delirium and exposure to psychoactive medications in critically ill adults: A multi-centre observational study. *J Crit Care* 2017;42:268-74. doi:10.1016/j.jccr.2017.08.003
 - 71 van den Boogaard M, Pickkers P, Slooter AJ, et al. Development and validation of PRE-DELIRIC (PREdiction of DELIRium in ICU patients) delirium prediction model for intensive care patients: observational multicentre study. *BMJ* 2012;344:e420. doi:10.1136/bmj.e420
 - 72 Serafim RB, Dutra MF, Saddy F, et al. Delirium in postoperative nonventilated intensive care patients: risk factors and outcomes. *Ann Intensive Care* 2012;2:51. doi:10.1186/2110-5820-2-51
 - 73 Colombo R, Corona A, Praga F, et al. A reorientation strategy for reducing delirium in the critically ill. Results of an interventional study. *Minerva Anestesiol* 2012;78:1026-33.
 - 74 Angles EM, Robinson TN, Biffi WL, et al. Risk factors for delirium after major trauma. *Am J Surg* 2008;196:864-9. discussion 869-70. doi:10.1016/j.amjsurg.2008.07.037
 - 75 Ouimet S, Kavanagh BP, Gottfried SB, Skrobik Y. Incidence, risk factors and consequences of ICU delirium. *Intensive Care Med* 2007;33:66-73. doi:10.1007/s00134-006-0399-8
 - 76 Dubois MJ, Bergeron N, Dumont M, Dial S, Skrobik Y. Delirium in an intensive care unit: a study of risk factors. *Intensive Care Med* 2001;27:1297-304. doi:10.1007/s001340101017
 - 77 Yoshitaka S, Egi M, Morimatsu H, Kanazawa T, Toda Y, Morita K. Perioperative plasma melatonin concentration in postoperative critically ill patients: its association with delirium. *J Crit Care* 2013;28:236-42. doi:10.1016/j.jccr.2012.11.004
 - 78 Guillaumondegui OD, Richards JE, Ely EW, et al. Does hypoxia affect intensive care unit delirium or long-term cognitive impairment after multiple trauma without intracranial hemorrhage? *J Trauma* 2011;70:910-5. doi:10.1097/TA.0b013e3182114f18
 - 79 Shi CM, Wang DX, Chen KS, Gu XE. Incidence and risk factors of delirium in critically ill patients after non-cardiac surgery. *Chin Med J (Engl)* 2010;123:993-9.
 - 80 Van Rompaey B, Elseviers MM, Schuurmans MJ, Shortridge-Baggett LM, Truijen S, Bossaert L. Risk factors for delirium in intensive care patients: a prospective cohort study. *Crit Care* 2009;13:R77. doi:10.1186/cc7892
 - 81 Shehabi Y, Chan L, Kadiman S, et al. Sedation Practice in Intensive Care Evaluation (SPICE) Study Group investigators. Sedation depth and long-term mortality in mechanically ventilated critically ill adults: a prospective longitudinal multicentre cohort study. *Intensive Care Med* 2013;39:910-8. doi:10.1007/s00134-013-2830-2
 - 82 Ruokonen E, Parviainen I, Jakob SM, et al. "Dexmedetomidine for Continuous Sedation" Investigators. Dexmedetomidine versus propofol/midazolam for long-term sedation during mechanical ventilation. *Intensive Care Med* 2009;35:282-90. doi:10.1007/s00134-008-1296-0
 - 83 Riker RR, Shehabi Y, Bokesch PM, et al. SEDCOM (Safety and Efficacy of Dexmedetomidine Compared With Midazolam) Study Group. Dexmedetomidine vs midazolam for sedation of critically ill patients: a randomized trial. *JAMA* 2009;301:489-99. doi:10.1001/jama.2009.56
 - 84 Pandharipande PP, Pun BT, Herr DL, et al. Effect of sedation with dexmedetomidine vs lorazepam on acute brain dysfunction in mechanically ventilated patients: the MENDS randomized controlled trial. *JAMA* 2007;298:2644-53. doi:10.1001/jama.298.22.2644
 - 85 Krewulak KD, Stelfox HT, Ely EW, Fiest KM. Risk factors and outcomes among delirium subtypes in adult ICUs: A systematic review. *J Crit Care* 2020;56:257-64. doi:10.1016/j.jccr.2020.01.017
 - 86 Schweickert WD, Pohlman MC, Pohlman AS, et al. Early physical and occupational therapy in mechanically ventilated, critically ill patients: a randomised controlled trial. *Lancet* 2009;373:1874-82. doi:10.1016/S0140-6736(09)60658-9
 - 87 Schaller SJ, Anstey M, Blobner M, et al. International Early SOMS-guided Mobilization Research Initiative. Early, goal-directed mobilisation in the surgical intensive care unit: a randomised controlled trial. *Lancet* 2016;388:1377-88. doi:10.1016/S0140-6736(16)31637-3
 - 88 Schaller SJ, Scheffenbichler FT, Bose S, et al. Influence of the initial level of consciousness on early, goal-directed mobilization: a post hoc analysis. *Intensive Care Med* 2019;45:201-10. doi:10.1007/s00134-019-05528-x
 - 89 Girard TD, Ely EW, MIND-USA Investigators. Haloperidol and Ziprasidone for Treatment of Delirium in Critical Illness. Reply. *N Engl J Med* 2019;380:1779-80.
 - 90 Nikoobir R, Neufeld KJ, Oh ES, et al. Antipsychotics for Treating Delirium in Hospitalized Adults: A Systematic Review. *Ann Intern Med* 2019;171:485-95. doi:10.7326/M19-1860
 - 91 Shehabi Y, Howe BD, Bellomo R, et al. ANZICS Clinical Trials Group and the SPICE III Investigators. Early Sedation with Dexmedetomidine in Critically Ill Patients. *N Engl J Med* 2019;380:2506-17. doi:10.1056/NEJMoa1904710
 - 92 Reade MC, Eastwood GM, Bellomo R, et al. DAHLIA Investigators, Australian and New Zealand Intensive Care Society Clinical Trials Group. Effect of Dexmedetomidine Added to Standard Care on Ventilator-Free Time in Patients With Agitated Delirium: A Randomized Clinical Trial. *JAMA* 2016;315:1460-8. doi:10.1001/jama.2016.2707
 - 93 Subramaniam B, Shankar P, Shaefi S, et al. Effect of Intravenous Acetaminophen vs Placebo Combined With Propofol or Dexmedetomidine on Postoperative Delirium Among Older Patients Following Cardiac Surgery: The DEXACET Randomized Clinical Trial. *JAMA* 2019;321:686-96. doi:10.1001/jama.2019.0234
 - 94 Alam A, Puri NV. Inefficacy of antipsychotics in treatment of delirium and agitation in two cases of bickerstaff brainstem encephalitis. *J Neuropsychiatry Clin Neurosci* 2014;26:176-8. doi:10.1176/appi.neuropsych.13040078
 - 95 Bourgeois JA, Koike AK, Simmons JE, Telles S, Eggleston C. Adjunctive valproic acid for delirium and/or agitation on a consultation-liaison service: a report of six cases. *J Neuropsychiatry Clin Neurosci* 2005;17:232-8. doi:10.1176/jnp.17.2.232
 - 96 Sher Y, Miller AC, Lolak S, Ament A, Maldonado JR. Adjunctive valproic acid in management-refractory hyperactive delirium: A case series and rationale. *J Neuropsychiatry Clin Neurosci* 2015;27:365-70. doi:10.1176/appi.neuropsych.14080190
 - 97 Subedi B, Louzon PR, Masood A, Recksieck S. Prolonged Refractory ICU Delirium Successfully Treated With Valproic Acid: Case Report and Literature Review. *J Pharm Pract* 2020;897190020953017.
 - 98 Gagnon DJ, Fontaine GV, Smith KE, et al. Valproate for agitation in critically ill patients: A retrospective study. *J Crit Care* 2017;37:119-25. doi:10.1016/j.jccr.2016.09.006
 - 99 Crowley KE, Urben L, Hacobian G, Geiger KL. Valproic Acid for the Management of Agitation and Delirium in the Intensive Care Setting:

- A Retrospective Analysis. *Clin Ther* 2020;42:e65-73. doi:10.1016/j.clinthera.2020.02.007
- 100 Quinn NJ, Hohlfelder B, Wanek MR, Duggal A, Torbic H. Prescribing Practices of Valproic Acid for Agitation and Delirium in the Intensive Care Unit. *Ann Pharmacother* 2021;55:311-7. doi:10.1177/1060028020947173
 - 101 Gagnon DJ, Fontaine GV, Riker RR, Fraser GL. Repurposing Valproate, Enteral Clonidine, and Phenobarbital for Comfort in Adult ICU Patients: A Literature Review with Practical Considerations. *Pharmacotherapy* 2017;37:1309-21. doi:10.1002/phar.2017
 - 102 Blyth BJ, Farhavar A, Gee C, et al. Validation of serum markers for blood-brain barrier disruption in traumatic brain injury. *J Neurotrauma* 2009;26:1497-507. doi:10.1089/neu.2008.0738
 - 103 van Munster BC, Korse CM, de Rooij SE, Bonfrer JM, Zwiderman AH, Korevaar JC. Markers of cerebral damage during delirium in elderly patients with hip fracture. *BMC Neurol* 2009;9:21. doi:10.1186/1471-2377-9-21
 - 104 Hughes CG, Morandi A, Girard TD, et al. Association between endothelial dysfunction and acute brain dysfunction during critical illness. *Anesthesiology* 2013;118:631-9. doi:10.1097/ALN.0b013e31827bd193
 - 105 Khan BA, Perkins AJ, Prasad NK, et al. Biomarkers of Delirium Duration and Delirium Severity in the ICU. *Crit Care Med* 2020;48:353-61. doi:10.1097/CCM.0000000000004139
 - 106 McGrane S, Girard TD, Thompson JL, et al. Procalcitonin and C-reactive protein levels at admission as predictors of duration of acute brain dysfunction in critically ill patients. *Crit Care* 2011;15:R78. doi:10.1186/cc10070
 - 107 Grandi C, Tomasi CD, Fernandes K, et al. Brain-derived neurotrophic factor and neuron-specific enolase, but not S100 β , levels are associated to the occurrence of delirium in intensive care unit patients. *J Crit Care* 2011;26:133-7. doi:10.1016/j.jcrc.2010.10.006
 - 108 van Munster BC, Bisschop PH, Zwiderman AH, et al. Cortisol, interleukins and S100B in delirium in the elderly. *Brain Cogn* 2010;74:18-23. doi:10.1016/j.bandc.2010.05.010
 - 109 Anderson BJ, Reilly JP, Shashaty MGS, et al. Admission plasma levels of the neuronal injury marker neuron-specific enolase are associated with mortality and delirium in sepsis. *J Crit Care* 2016;36:18-23. doi:10.1016/j.jcrc.2016.06.012
 - 110 Hughes CG, Pandharipande PP, Thompson JL, et al. Endothelial Activation and Blood-Brain Barrier Injury as Risk Factors for Delirium in Critically Ill Patients. *Crit Care Med* 2016;44:e809-17. doi:10.1097/CCM.0000000000001739
 - 111 Girard TD, Ware LB, Bernard GR, et al. Associations of markers of inflammation and coagulation with delirium during critical illness. *Intensive Care Med* 2012;38:1965-73. doi:10.1007/s00134-012-2678-x
 - 112 Simons KS, van den Boogaard M, Hendriksen E, et al. Temporal biomarker profiles and their association with ICU acquired delirium: a cohort study. *Crit Care* 2018;22:137. doi:10.1186/s13054-018-2054-5
 - 113 Morandi A, Gunther ML, Vasilevskis EE, et al. Neuroimaging in delirious intensive care unit patients: a preliminary case series report. *Psychiatry (Edmont)* 2010;7:28-33.
 - 114 Morandi A, Rogers BP, Gunther ML, et al. VISIONS Investigation, ViSualizing Icu SurvivOrs Neuroradiological Sequelae. The relationship between delirium duration, white matter integrity, and cognitive impairment in intensive care unit survivors as determined by diffusion tensor imaging: the VISIONS prospective cohort magnetic resonance imaging study. *Crit Care Med* 2012;40:2182-9. doi:10.1097/CCM.0b013e318250acdc
 - 115 Kimchi EY, Neelagiri A, Whitt W, et al. Clinical EEG slowing correlates with delirium severity and predicts poor clinical outcomes. *Neurology* 2019;93:e1260-71. doi:10.1212/WNL.00000000000008164
 - 116 Palanca BJA, Wildes TS, Ju YS, Ching S, Avidan MS. Electroencephalography and delirium in the postoperative period. *Br J Anaesth* 2017;119:294-307. doi:10.1093/bja/aew475
 - 117 van der Kooi AW, Zaal IJ, Klijn FA, et al. Delirium detection using EEG: what and how to measure. *Chest* 2015;147:94-101. doi:10.1378/chest.13-3050
 - 118 Jacobson SA, Leuchter AF, Walter DO. Conventional and quantitative EEG in the diagnosis of delirium among the elderly. *J Neurol Neurosurg Psychiatry* 1993;56:153-8. doi:10.1136/jnnp.56.2.153
 - 119 Numan T, van den Boogaard M, Kamper AM, Rood PJT, Peelen LM, Slooter AJC, Dutch Delirium Detection Study Group. Delirium detection using relative delta power based on 1-minute single-channel EEG: a multicentre study. *Br J Anaesth* 2019;122:60-8. doi:10.1016/j.bja.2018.08.021
 - 120 Norman BC, Jackson JC, Graves JA, et al. Employment Outcomes After Critical Illness: An Analysis of the Bringing to Light the Risk Factors and Incidence of Neuropsychological Dysfunction in ICU Survivors Cohort. *Crit Care Med* 2016;44:2003-9. doi:10.1097/CCM.0000000000001849
 - 121 Estrup S, Kjer CKW, Vilhelmsen F, Poulsen LM, Gøgenur I, Mathiesen O. Cognitive Function 3 and 12 Months After ICU Discharge-A Prospective Cohort Study. *Crit Care Med* 2018;46:e1121-7. doi:10.1097/CCM.0000000000003391
 - 122 Denke C, Balzer F, Menk M, et al. Long-term sequelae of acute respiratory distress syndrome caused by severe community-acquired pneumonia: Delirium-associated cognitive impairment and post-traumatic stress disorder. *J Int Med Res* 2018;46:2265-83. doi:10.1177/0300060518762040
 - 123 Calsavara AJC, Costa PA, Nobre V, Teixeira AL. Factors Associated With Short and Long Term Cognitive Changes in Patients With Sepsis. *Sci Rep* 2018;8:4509. doi:10.1038/s41598-018-22754-3
 - 124 de Azevedo JR, Montenegro WS, Rodrigues DP, et al. Long-term cognitive outcomes among unselected ventilated and non-ventilated ICU patients. *J Intensive Care* 2017;5:18. doi:10.1186/s40560-017-0213-4
 - 125 Honarmand K, Lalli RS, Priestard F, et al. Natural History of Cognitive Impairment in Critical Illness Survivors. A Systematic Review. *Am J Respir Crit Care Med* 2020;202:193-201. doi:10.1164/rccm.201904-0816CI
 - 126 Wilcox ME, Brummel NE, Archer K, Ely EW, Jackson JC, Hopkins RO. Cognitive dysfunction in ICU patients: risk factors, predictors, and rehabilitation interventions. *Crit Care Med* 2013;41(Suppl 1):S81-98. doi:10.1097/CCM.0b013e3182a16946
 - 127 Hopkins RO, Weaver LK, Collingridge D, Parkinson RB, Chan KJ, Orme JF Jr. Two-year cognitive, emotional, and quality-of-life outcomes in acute respiratory distress syndrome. *Am J Respir Crit Care Med* 2005;171:340-7. doi:10.1164/rccm.200406-7630C
 - 128 Jackson JC, Archer KR, Bauer R, et al. A prospective investigation of long-term cognitive impairment and psychological distress in moderately versus severely injured trauma intensive care unit survivors without intracranial hemorrhage. *J Trauma* 2011;71:860-6. doi:10.1097/TA.0b013e3182151961
 - 129 Jackson JC, Girard TD, Gordon SM, et al. Long-term cognitive and psychological outcomes in the awakening and breathing controlled trial. *Am J Respir Crit Care Med* 2010;182:183-91. doi:10.1164/rccm.200903-0442OC
 - 130 Needham DM, Davidson J, Cohen H, et al. Improving long-term outcomes after discharge from intensive care unit: report from a stakeholders' conference. *Crit Care Med* 2012;40:502-9. doi:10.1097/CCM.0b013e318232da75
 - 131 Ehlenbach WJ, Hough CL, Crane PK, et al. Association between acute care and critical illness hospitalization and cognitive function in older adults. *JAMA* 2010;303:763-70. doi:10.1001/jama.2010.167
 - 132 Iwashyna TJ, Ely EW, Smith DM, Langa KM. Long-term cognitive impairment and functional disability among survivors of severe sepsis. *JAMA* 2010;304:1787-94. doi:10.1001/jama.2010.1553
 - 133 Geense WW, van den Boogaard M, Peters MAA, et al. Physical, Mental, and Cognitive Health Status of ICU Survivors Before ICU Admission: A Cohort Study. *Crit Care Med* 2020;48:1271-9. doi:10.1097/CCM.0000000000004443
 - 134 Teeters DA, Moua T, Li G, et al. Mild Cognitive Impairment and Risk of Critical Illness. *Crit Care Med* 2016;44:2045-51. doi:10.1097/CCM.0000000000001842
 - 135 Ferrante LE, Murphy TE, Gahbauer EA, Leo-Summers LS, Pisani MA, Gill TM. Pre-Intensive Care Unit Cognitive Status, Subsequent Disability, and New Nursing Home Admission among Critically Ill Older Adults. *Ann Am Thorac Soc* 2018;15:622-9. doi:10.1513/AnnalsATS.201709-7020C
 - 136 Gunther ML, Morandi A, Krauskopf E, et al. VISIONS Investigation, ViSualizing Icu SurvivOrs Neuroradiological Sequelae. The association between brain volumes, delirium duration, and cognitive outcomes in intensive care unit survivors: the VISIONS cohort magnetic resonance imaging study. *Crit Care Med* 2012;40:2022-32. doi:10.1097/CCM.0b013e318250acdc
 - 137 van den Boogaard M, Schoonhoven L, Evers AW, van der Hoeven JG, van Achterberg T, Pickkers P. Delirium in critically ill patients: impact on long-term health-related quality of life and cognitive functioning. *Crit Care Med* 2012;40:112-8. doi:10.1097/CCM.0b013e31822e9fc9
 - 138 Brummel NE, Girard TD, Ely EW, et al. Feasibility and safety of early combined cognitive and physical therapy for critically ill medical and surgical patients: the Activity and Cognitive Therapy in ICU (ACT-ICU) trial. *Intensive Care Med* 2014;40:370-9. doi:10.1007/s00134-013-3136-0
 - 139 Jackson JC, Morandi A, Girard TD, et al. ViSualizing Icu SurvivOrs Neuroradiological Sequelae (VISIONS) Investigation. Functional brain imaging in survivors of critical illness: A prospective feasibility study and exploration of the association between delirium and brain activation patterns. *J Crit Care* 2015;30:653.e1-7. doi:10.1016/j.jcrc.2015.01.017
 - 140 Wolters AE, Peelen LM, Veldhuijzen DS, et al. Long-Term Self-Reported Cognitive Problems After Delirium in the Intensive Care Unit and the Effect of Systemic Inflammation. *J Am Geriatr Soc* 2017;65:786-91. doi:10.1111/jgs.14660

- 141 Brück E, Schandl A, Bottai M, Sackey P. The impact of sepsis, delirium, and psychological distress on self-rated cognitive function in ICU survivors—a prospective cohort study. *J Intensive Care* 2018;6:2. doi:10.1186/s40560-017-0272-6
- 142 Mitchell ML, Shum DHK, Mihalá G, Murfield JE, Aitken LM. Long-term cognitive impairment and delirium in intensive care: A prospective cohort study. *Aust Crit Care* 2018;31:204–11. doi:10.1016/j.aucc.2017.07.002
- 143 Zhang DF, Su X, Meng ZT, et al. Impact of Dexmedetomidine on Long-term Outcomes After Noncardiac Surgery in Elderly: 3-Year Follow-up of a Randomized Controlled Trial. *Ann Surg* 2019;270:356–63. doi:10.1097/SLA.0000000000002801
- 144 Schulte PJ, Warner DO, Martin DP, et al. Association Between Critical Care Admissions and Cognitive Trajectories in Older Adults. *Crit Care Med* 2019;47:1116–24. doi:10.1097/CCM.0000000000003829
- 145 Bulic D, Bennett M, Georgousopoulou EN, et al. Cognitive and psychosocial outcomes of mechanically ventilated intensive care patients with and without delirium. *Ann Intensive Care* 2020;10:104. doi:10.1186/s13613-020-00723-2
- 146 Sakuramoto H, Subrina J, Unoki T, Mizutani T, Komatsu H. Severity of delirium in the ICU is associated with short term cognitive impairment. A prospective cohort study. *Intensive Crit Care Nurs* 2015;31:250–7. doi:10.1016/j.iccn.2015.01.001
- 147 Khan BA, Zawahir M, Campbell NL, Boustani MA. Biomarkers for delirium—a review. *J Am Geriatr Soc* 2011;59(Suppl 2):S256–61. doi:10.1111/j.1532-5415.2011.03702.x
- 148 Hall RJ, Shenkin SD, MacLullich AMJ. A systematic literature review of cerebrospinal fluid biomarkers in delirium. *Dement Geriatr Cogn Disord* 2011;32:79–93. doi:10.1159/000330757
- 149 van Gool WA, van de Beek D, Eikelenboom P. Systemic infection and delirium: when cytokines and acetylcholine collide. *Lancet* 2010;375:773–5. doi:10.1016/S0140-6736(09)61158-2
- 150 Li Y, Yu ZX, Ji MS, et al. A Pilot Study of the Use of Dexmedetomidine for the Control of Delirium by Reducing the Serum Concentrations of Brain-Derived Neurotrophic Factor, Neuron-Specific Enolase, and S100B in Polytrauma Patients. *J Intensive Care Med* 2019;34:674–81. doi:10.1177/0885066617710643
- 151 Brummel NE, Hughes CG, Thompson JL, et al. Inflammation and Coagulation during Critical Illness and Long-Term Cognitive Impairment and Disability. *Am J Respir Crit Care Med* 2021;203:699–706. doi:10.1164/rccm.201912-24490C
- 152 Porhomayon J, Joude P, Adlparvar G, El-Solh AA, Nader ND. The Impact of High Versus Low Sedation Dosing Strategy on Cognitive Dysfunction in Survivors of Intensive Care Units: A Systematic Review and Meta-Analysis. *J Cardiovasc Thorac Res* 2015;7:43–8. doi:10.15171/jcvtr.2015.10
- 153 Inoue SK, Ferrucci L. Elucidating the pathophysiology of delirium and the interrelationship of delirium and dementia. *J Gerontol A Biol Sci Med Sci* 2006;61:1277–80. doi:10.1093/gerona/61.12.1277
- 154 Woon FL, Dunn CB, Hopkins RO. Predicting cognitive sequelae in survivors of critical illness with cognitive screening tests. *Am J Respir Crit Care Med* 2012;186:333–40. doi:10.1164/rccm.201112-22610C
- 155 Haines KJ, Hibbert E, McPeake J, et al. Prediction Models for Physical, Cognitive, and Mental Health Impairments After Critical Illness: A Systematic Review and Critical Appraisal. *Crit Care Med* 2020;48:1871–80. doi:10.1097/CCM.00000000000004659
- 156 Nedergaard HK, Jensen HI, Toft P. Interventions to reduce cognitive impairments following critical illness: a topical systematic review. *Acta Anaesthesiol Scand* 2017;61:135–48. doi:10.1111/aas.12832
- 157 Kamdar BB, King LM, Collop NA, et al. The effect of a quality improvement intervention on perceived sleep quality and cognition in a medical ICU. *Crit Care Med* 2013;41:800–9. doi:10.1097/CCM.0b013e3182746442
- 158 Needham DM, Colantuoni E, Dinglas VD, et al. Rosuvastatin versus placebo for delirium in intensive care and subsequent cognitive impairment in patients with sepsis-associated acute respiratory distress syndrome: an ancillary study to a randomised controlled trial. *Lancet Respir Med* 2016;4:203–12. doi:10.1016/S2213-2600(16)00005-9
- 159 Needham DM, Dinglas VD, Morris PE, et al. NIH NHLBI ARDS Network. Physical and cognitive performance of patients with acute lung injury 1 year after initial trophic versus full enteral feeding. EDEN trial follow-up. *Am J Respir Crit Care Med* 2013;188:567–76. doi:10.1164/rccm.201304-06510C
- 160 Mikkelsen ME, Christie JD, Lanken PN, et al. The adult respiratory distress syndrome cognitive outcomes study: long-term neuropsychological function in survivors of acute lung injury. *Am J Respir Crit Care Med* 2012;185:1307–15. doi:10.1164/rccm.201111-20250C
- 161 Jackson JC, Ely EW, Morey MC, et al. Cognitive and physical rehabilitation of intensive care unit survivors: results of the RETURN randomized controlled pilot investigation. *Crit Care Med* 2012;40:1088–97. doi:10.1097/CCM.0b013e3182373115
- 162 Wang S, Hammes J, Khan S, et al. Improving Recovery and Outcomes Every Day after the ICU (IMPROVE): study protocol for a randomized controlled trial. *Trials* 2018;19:196. doi:10.1186/s13063-018-2569-8
- 163 Fonsmark L, Hein L, Nibroe H, et al. Danish Society of Intensive Care Medicine (DSIT) and the Danish Society of Anesthesiology and Intensive Care Medicine (DASAIM). Danish national sedation strategy. Targeted therapy of discomfort associated with critical illness. *Dan Med J* 2015;62:C5052.
- 164 Baron R, Binder A, Biniek R, et al. DAS-Taskforce 2015. Evidence and consensus based guideline for the management of delirium, analgesia, and sedation in intensive care medicine. Revision 2015 (DAS-Guideline 2015) - short version. *Ger Med Sci* 2015;13:Doc19.
- 165 Committee for the development of Japanese guidelines for the management of Pain, Agitation, and Delirium in intensive care unit, Japanese Society of Intensive Care Medicine. Japanese guidelines for the management of Pain, Agitation, and Delirium in intensive care unit (J-PAD). *Journal of the Japanese Society of Intensive Care Medicine* 2014;21:539–79. doi:10.3918/jsicm.21.539
- 166 Girardis M, Cantaroni C, Savoia G, Melotti R, Conti G. A critical appraisal of the quality of analgesia/sedation guidelines in critically ill patients. *Minerva Anestesiol* 2016;82:230–5.
- 167 Balas MC, Devlin JW, Verceles AC, Morris P, Ely EW. Adapting the ABCDEF Bundle to Meet the Needs of Patients Requiring Prolonged Mechanical Ventilation in the Long-Term Acute Care Hospital Setting: Historical Perspectives and Practical Implications. *Semin Respir Crit Care Med* 2016;37:119–35. doi:10.1055/s-0035-1570361
- 168 Balas MC, Burke WJ, Gannon D, et al. Implementing the awakening and breathing coordination, delirium monitoring/management, and early exercise/mobility bundle into everyday care: opportunities, challenges, and lessons learned for implementing the ICU Pain, Agitation, and Delirium Guidelines. *Crit Care Med* 2013;41(Suppl 1):S116–27. doi:10.1097/CCM.0b013e3182a17064
- 169 Balas MC, Pun BT, Pasero C, et al. Common Challenges to Effective ABCDEF Bundle Implementation: The ICU Liberation Campaign Experience. *Crit Care Nurse* 2019;39:46–60. doi:10.4037/ccn2019927
- 170 Stollings JL, Devlin JW, Pun BT, et al. Implementing the ABCDEF Bundle: Top 8 Questions Asked During the ICU Liberation ABCDEF Bundle Improvement Collaborative. *Crit Care Nurse* 2019;39:36–45. doi:10.4037/ccn2019981
- 171 Kram SL, DiBartolo MC, Hinderer K, Jones RA. Implementation of the ABCDE Bundle to Improve Patient Outcomes in the Intensive Care Unit in a Rural Community Hospital. *Dimens Crit Care Nurs* 2015;34:250–8. doi:10.1097/DCC.0000000000000129
- 172 Bounds M, Kram S, Speroni KG, et al. Effect of ABCDE Bundle Implementation on Prevalence of Delirium in Intensive Care Unit Patients. *Am J Crit Care* 2016;25:535–44. doi:10.4037/ajcc2016209
- 173 Hsieh SJ, Otusanya O, Gershengorn HB, et al. Staged Implementation of Awakening and Breathing, Coordination, Delirium Monitoring and Management, and Early Mobilization Bundle Improves Patient Outcomes and Reduces Hospital Costs. *Crit Care Med* 2019;47:885–93. doi:10.1097/CCM.0000000000003765
- 174 Heim M, Draheim R, Krupp A, et al. Evaluation of a Multidisciplinary Pain, Agitation, and Delirium Guideline in Mechanically Ventilated Critically Ill Adults. *Hosp Pharm* 2019;54:119–24. doi:10.1177/0018578718769570
- 175 Trogrlić Z, van der Jagt M, Lingsma H, et al. Improved Guideline Adherence and Reduced Brain Dysfunction After a Multicenter Multifaceted Implementation of ICU Delirium Guidelines in 3,930 Patients. *Crit Care Med* 2019;47:419–27. doi:10.1097/CCM.0000000000003596
- 176 Trogrlić Z, Ista E, Ponssen HH, et al. Attitudes, knowledge and practices concerning delirium: a survey among intensive care unit professionals. *Nurs Crit Care* 2017;22:133–40. doi:10.1111/nicc.12239
- 177 Maximous R, Miller F, Tan C, et al. Pain, agitation and delirium assessment and management in a community medical-surgical ICU: results from a prospective observational study and nurse survey. *BMJ Open Qual* 2018;7:e000413. doi:10.1136/bmjopen-2018-000413
- 178 Aparanji K, Kulkarni S, Metzke M, Schmudde Y, White P, Jaeger C. Quality improvement of delirium status communication and documentation for intensive care unit patients during daily multidisciplinary rounds. *BMJ Open Qual* 2018;7:e000239. doi:10.1136/bmjopen-2017-000239
- 179 Rood P, Huisman-de Waal G, Vermeulen H, Schoonhoven L, Pickers P, van den Boogaard M. Effect of organisational factors on the variation in incidence of delirium in intensive care unit patients: A systematic review and meta-regression analysis. *Aust Crit Care* 2018;31:180–7. doi:10.1016/j.aucc.2018.02.002
- 180 Esposito K, Nappo F, Marfella R, et al. Inflammatory cytokine concentrations are acutely increased by hyperglycemia in humans: role of oxidative stress. *Circulation* 2002;106:2067–72. doi:10.1161/01.CIR.0000034509.14906.AE